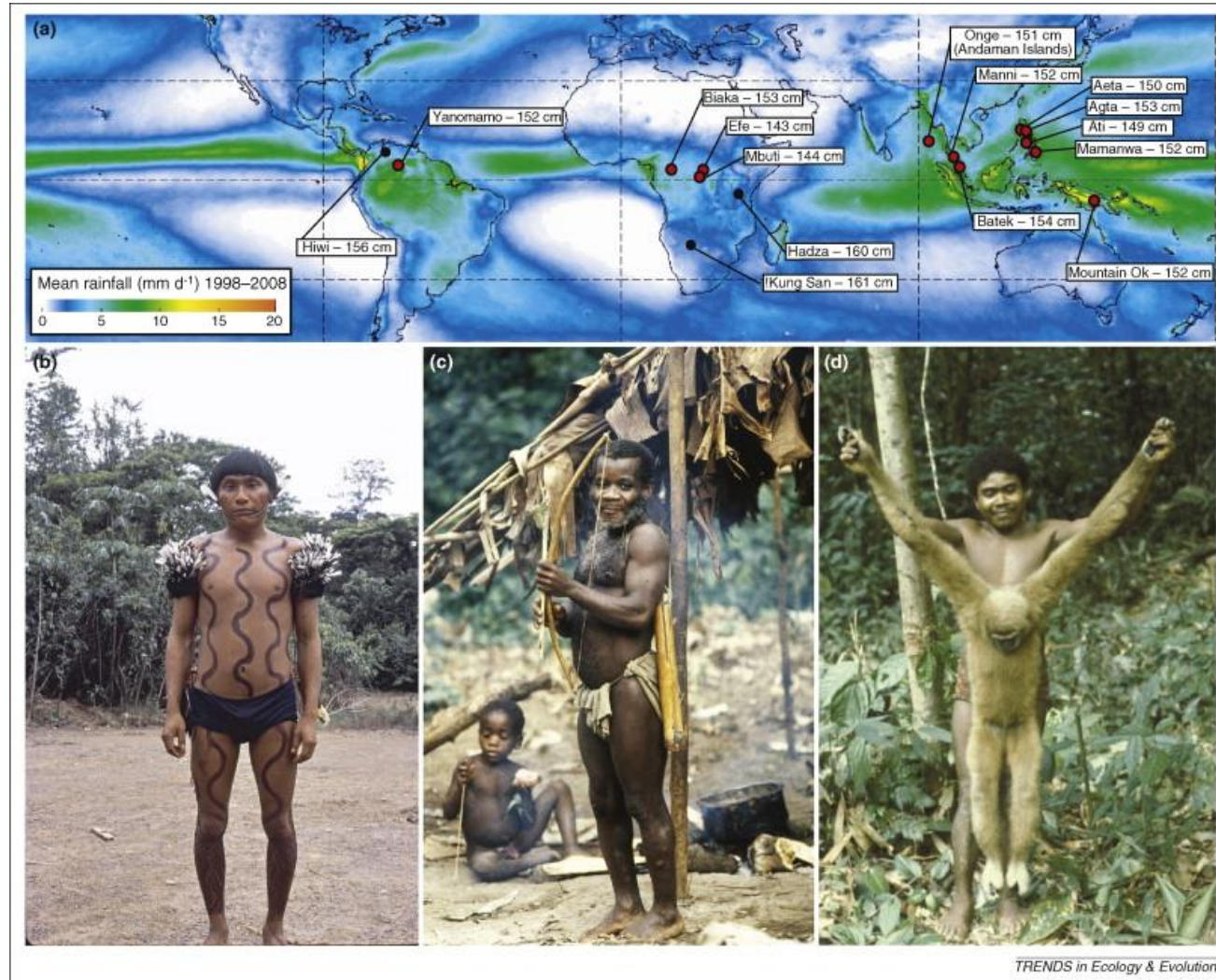


Interesting outliers to these general patterns



In groups, brainstorm potential drivers – whether proximate or ultimate! – of pygmy stature.

Choose one, and design a research question that you might use to inform a research project.

(Practice for your Research Project Design).

Competing explanations

Humidity?

Small size reduces heat generated.

But:

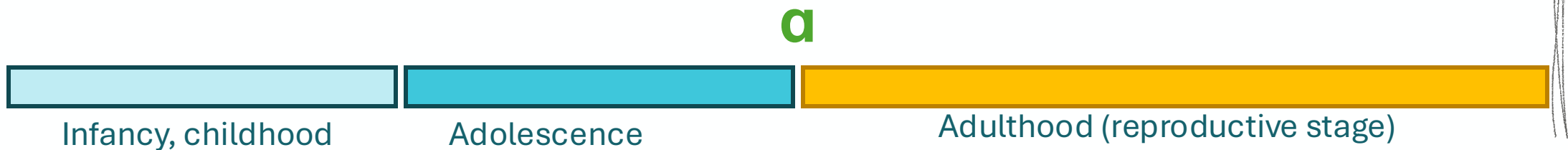
- Efe: low humidity
- Taller populations in PNG in humid environments



Extrinsic mortality and growth: Charnov's model

Low adult mortality

Death

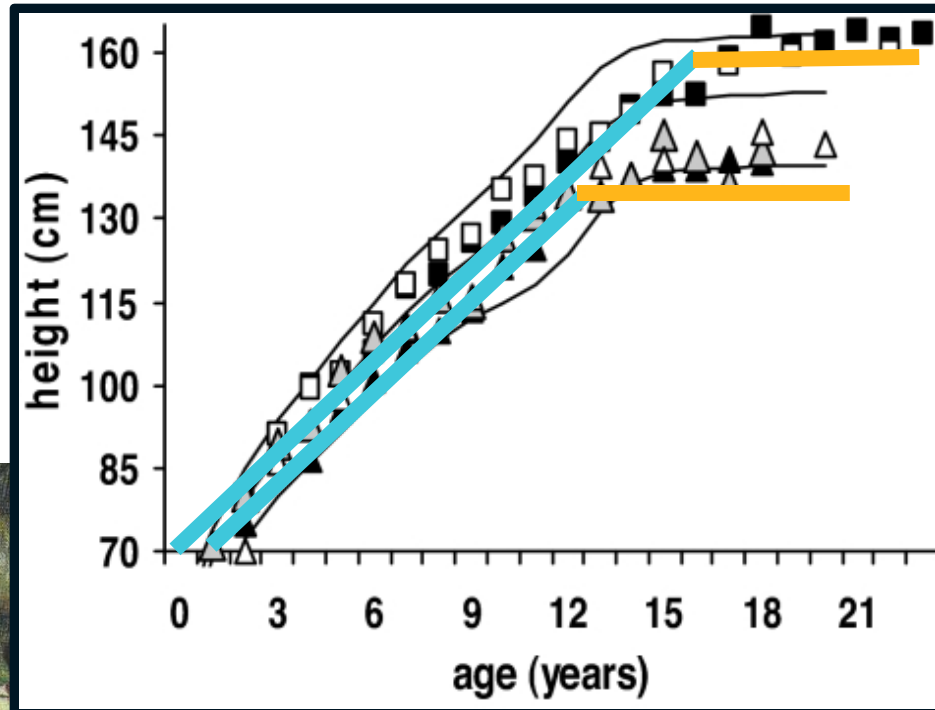


Given *adult mortalities*, selection sets α according to the tradeoff between the benefits of growing longer vs. reproducing sooner.

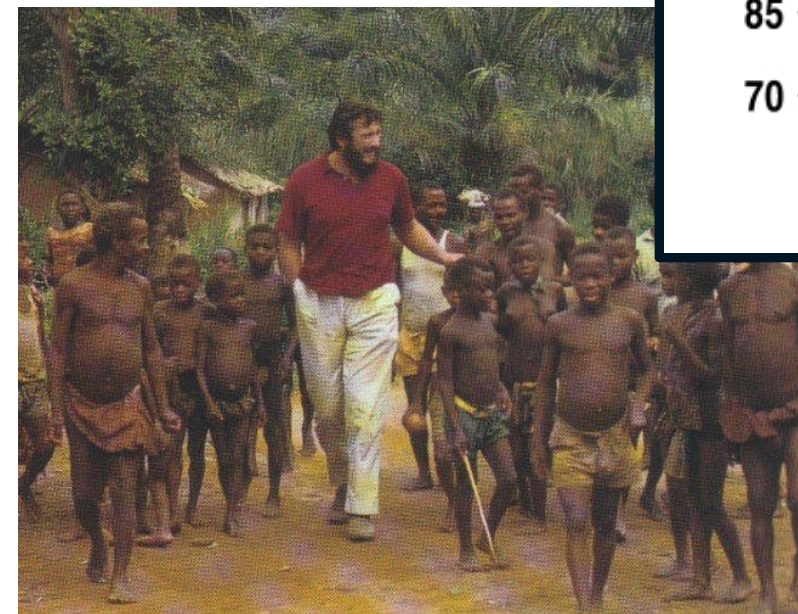
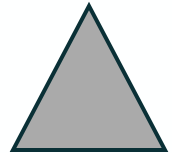
earlier adult mortality → switch to focus on reproduction earlier → Don't get as having more babies instead big, focus on



Extrinsic mortality and growth: Charnov's model applied to explain pygmy stature

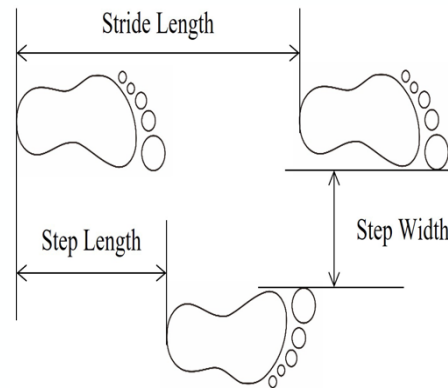
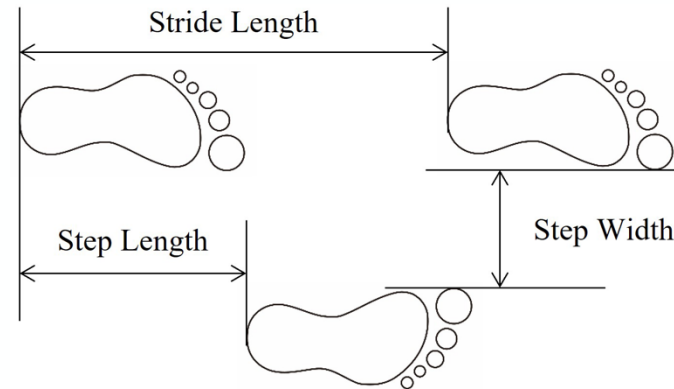
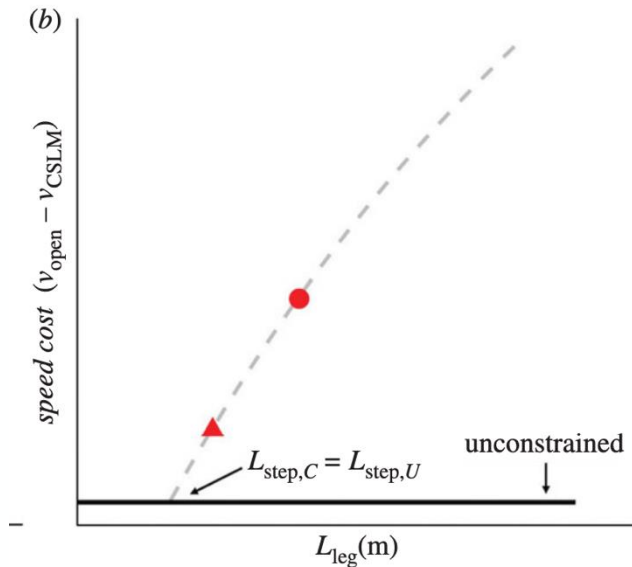


slope of
plot shows
rate of growth
Triangles - pygmy pops - squares - taller



Competing explanations

Constrained environment
like a miniforest - hard to
walk in a rain forest esp. if
you're tall



In unconstrained environments: lower cost per unit distance at a given speed, because fewer steps!

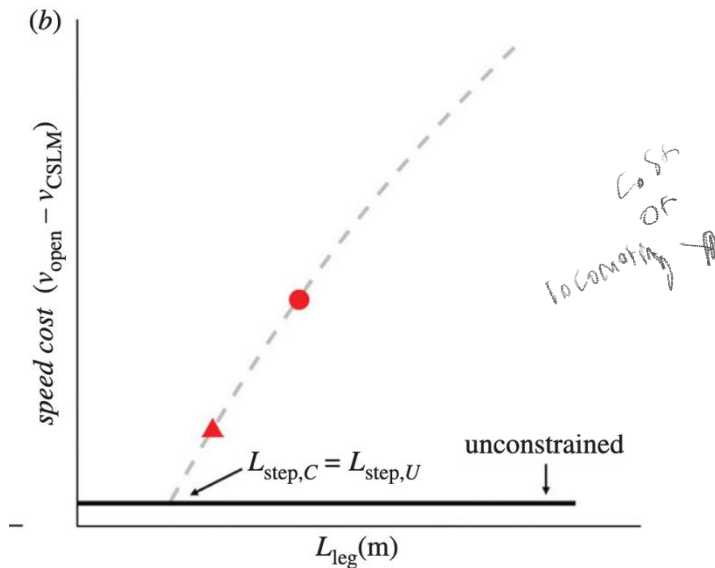


Expectation:

Cost of walking increases with leg length in constrained environments, because tall individuals need to walk more slowly than they would prefer!

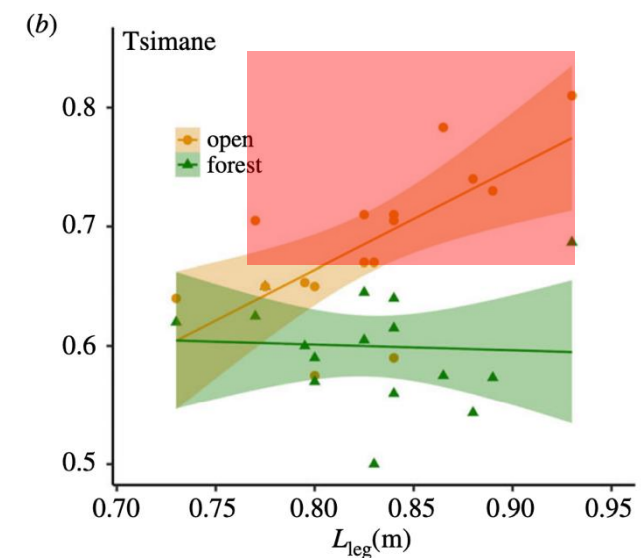
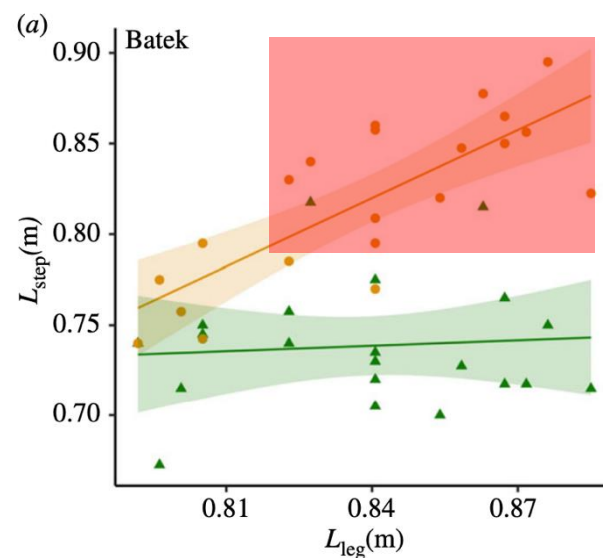
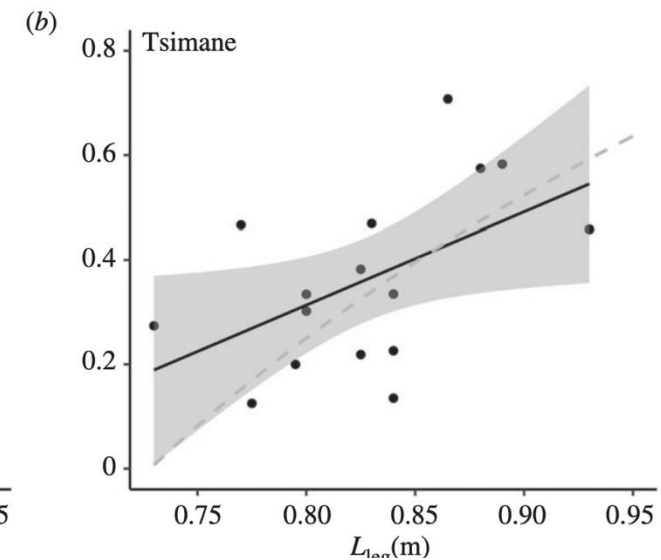
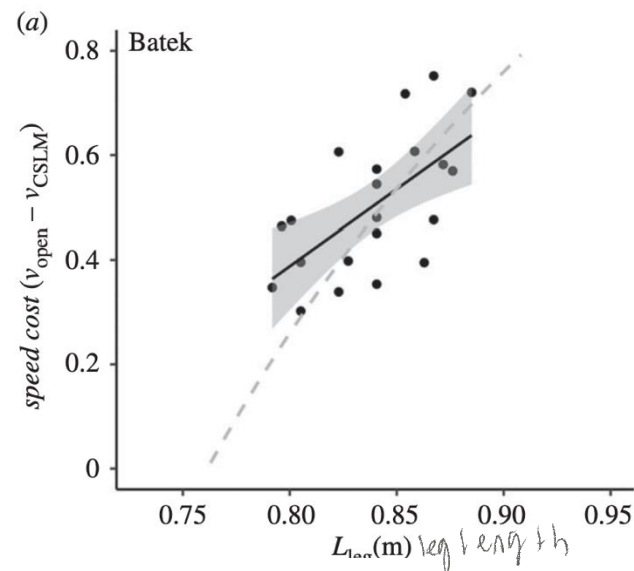
This costs energy

Competing explanations



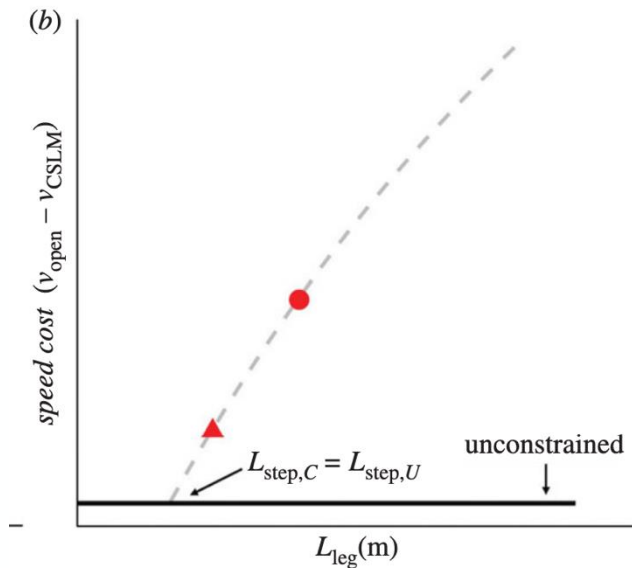
Expectation:

Cost of walking increases with leg length *in constrained environments*, because tall individuals need to walk more slowly than they would prefer!



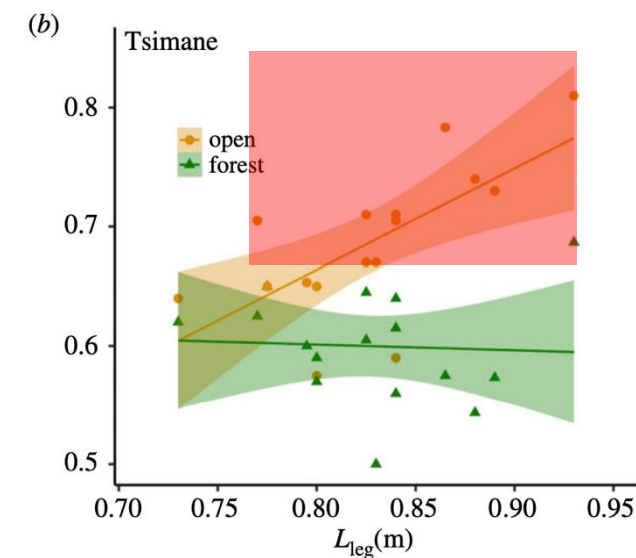
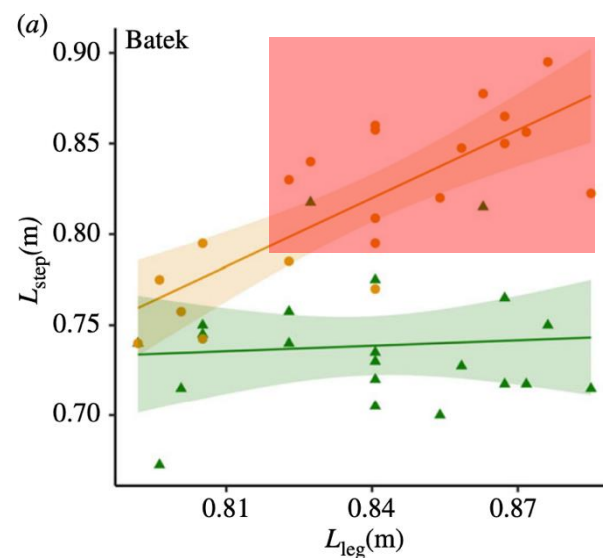
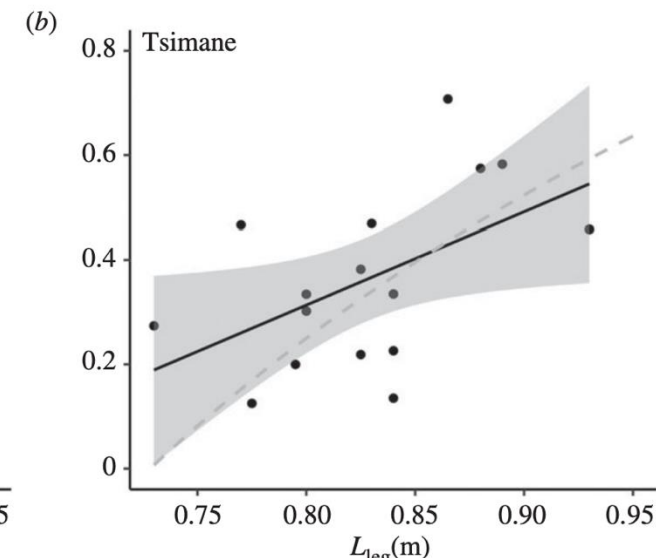
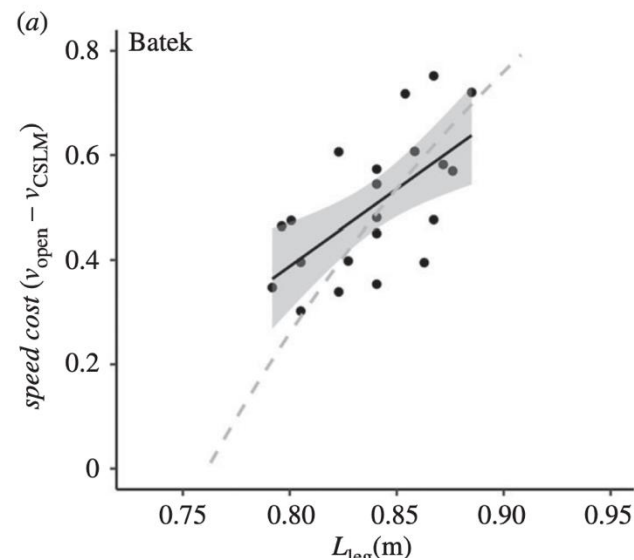
Competing explanations (or are they?)

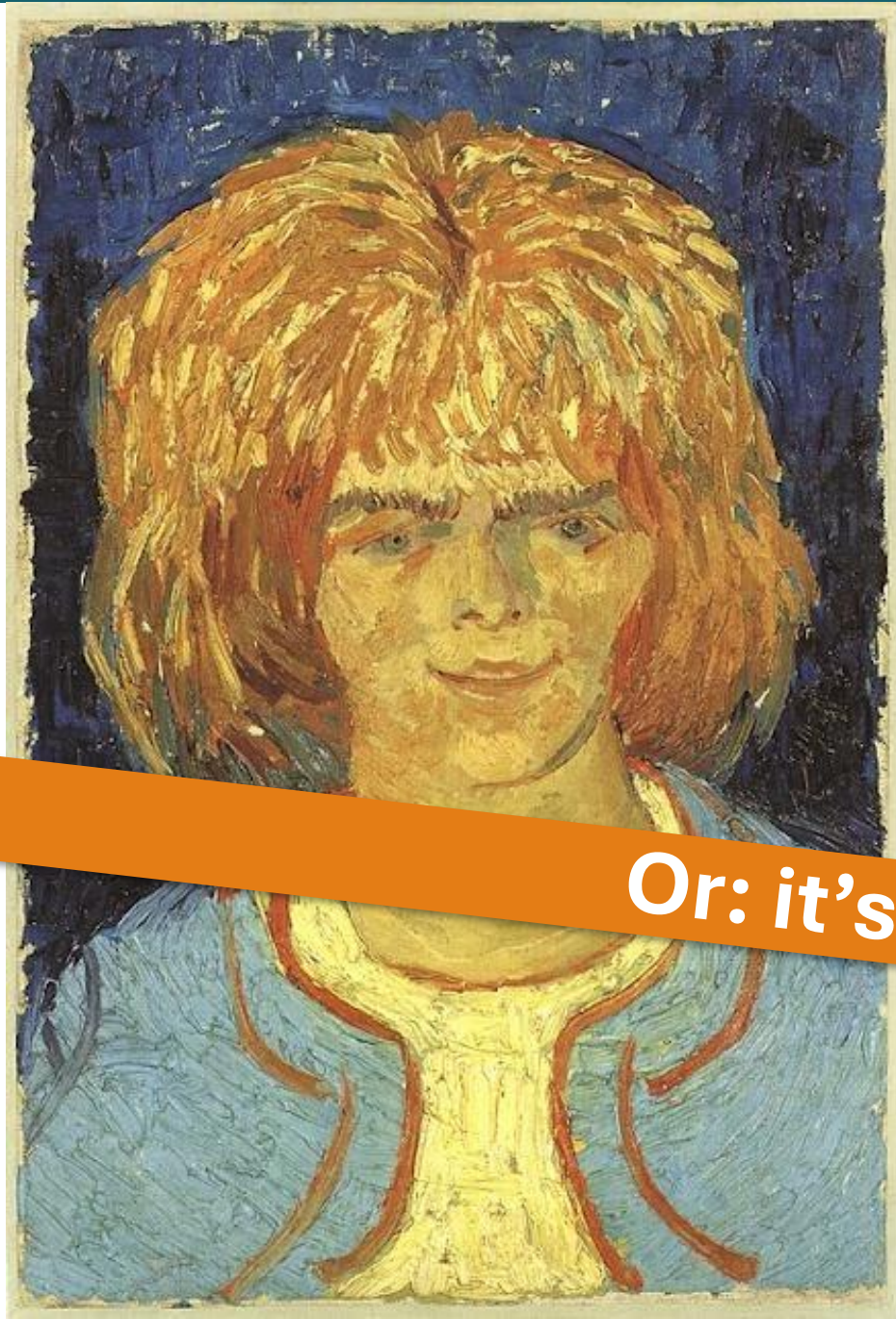
this is more of an approximate explanation
whereas growth rates are v1 finite



Expectation:

Cost of walking increases with leg length *in constrained environments*, because tall individuals need to walk more slowly than they would prefer!





Skin and hair diversity

Dr. Laura van Holstein

lavanho@emory.edu

Anthropology 215

Or: it's all the sun's fault



The naked ape

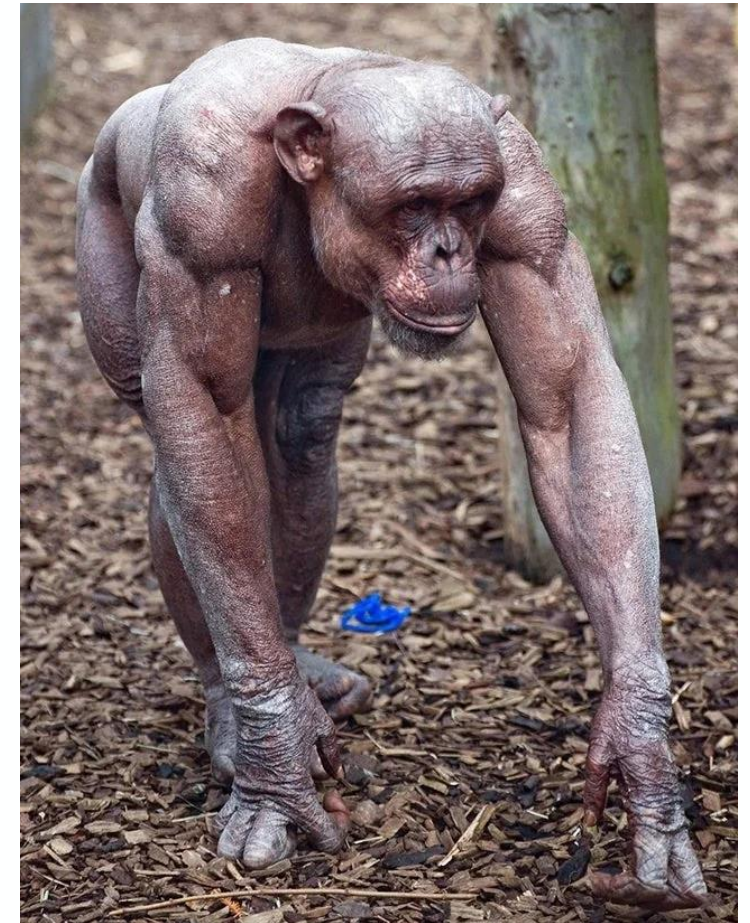




The *nearly* naked ape



The *nearly* naked ape
with exceptional skin colour
diversity



Colourful faces, but skin
underneath fur is light

- Why did humans lose their fur?
- Why do humans have scalp hair that is so exceptionally diverse in colour *and* structure?
- How is scalp hair colour and structure distributed across the world? Why?

- Why did humans lose their fur?
- Why do humans have skin that is so exceptionally diverse in colour?
- How is skin colour distributed across the world? Why?

The *nearly* naked ape
with exceptional skin colour
diversity

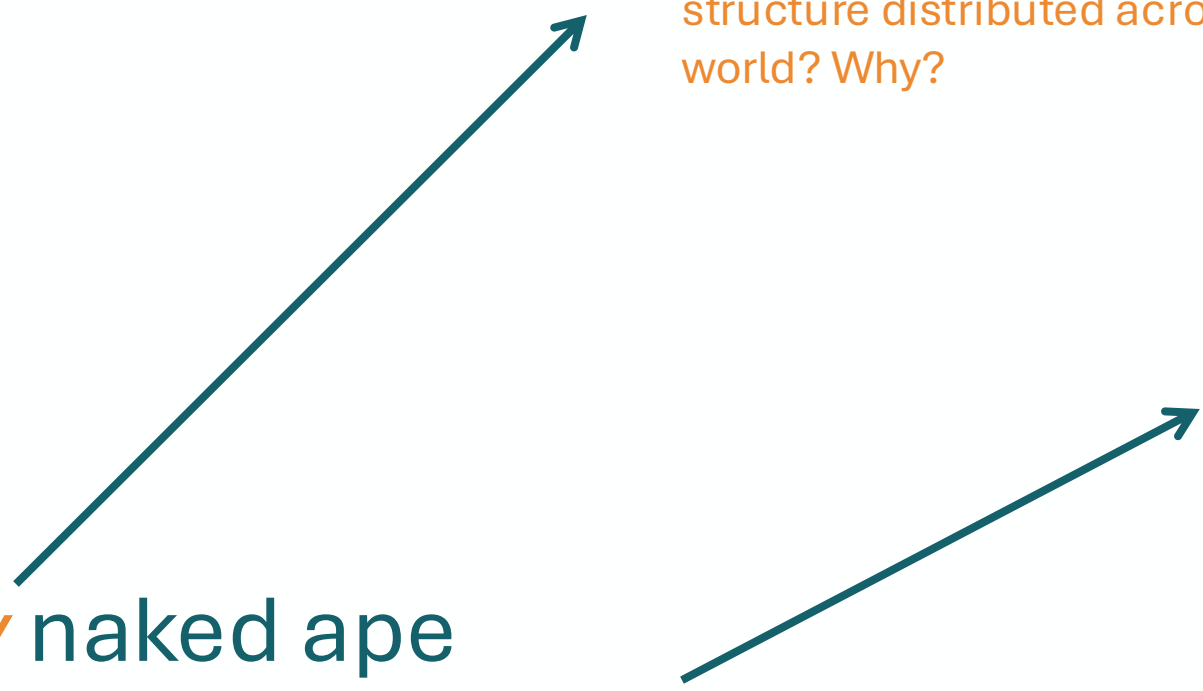


The diagram illustrates the relationship between human skin and hair diversity and evolutionary questions. It features a main text block at the bottom left, two bulleted lists of questions at the top right, and two arrows pointing from the main text to the lists. The first arrow points from 'nearly' to the first list, and the second arrow points from 'skin colour' to the second list. The words 'nearly' and 'skin colour' are highlighted in orange and green respectively, matching the color of the text in the lists they point to.

- Why did humans lose their fur?
- Why do humans have scalp hair that is so exceptionally diverse in colour *and* structure?
- How is scalp hair colour and structure distributed across the world? Why?

- Why did humans lose their fur?
- Why do humans have skin that is so exceptionally diverse in colour?
- How is skin colour distributed across the world? Why?

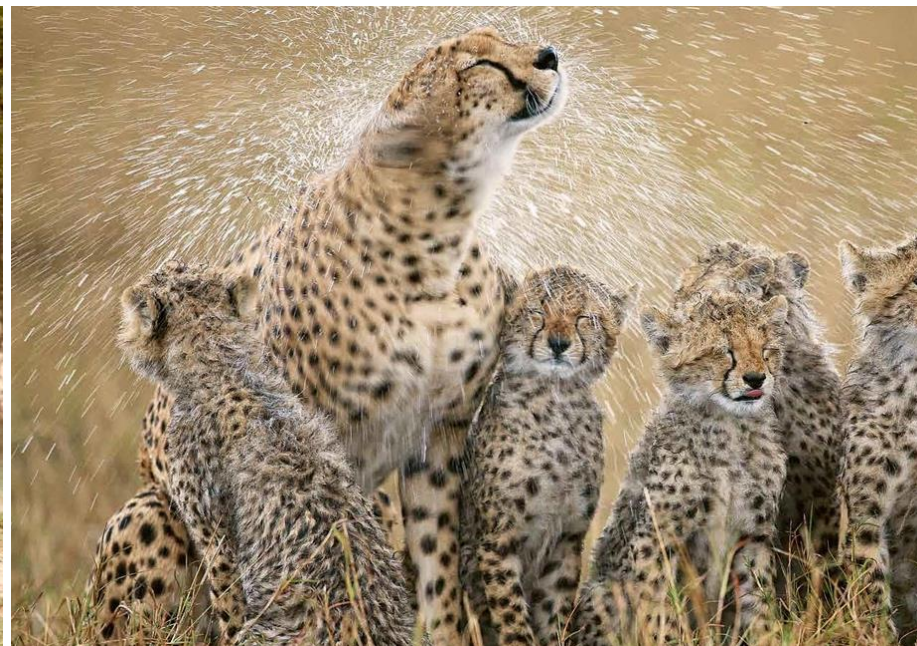
The *nearly* naked ape
with exceptional skin colour
diversity



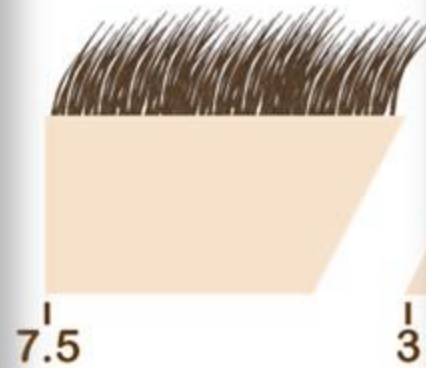
**Hair:**

- Insulates – from cold and heat
 - Protects against abrasion and UVR → UV radiation
 - Sheds water
 - Slows ingress of pathogens
- also useful in communication
↳ raising fur in displays of aggression etc

Humans lose all these
very adaptive functions

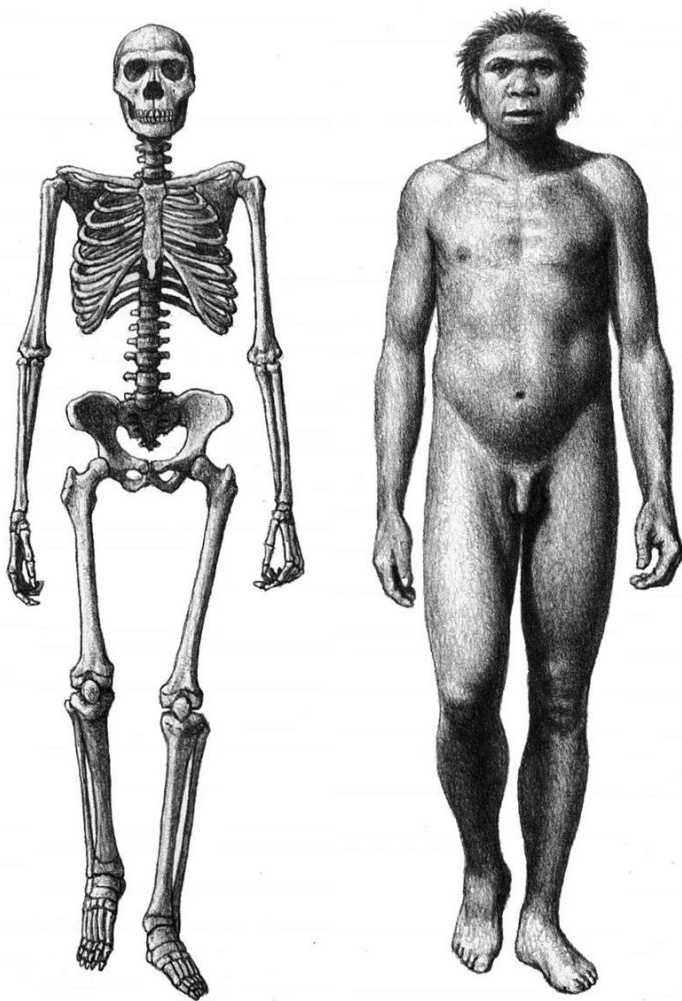


1. Why did humans *lose (most of)* their fur?

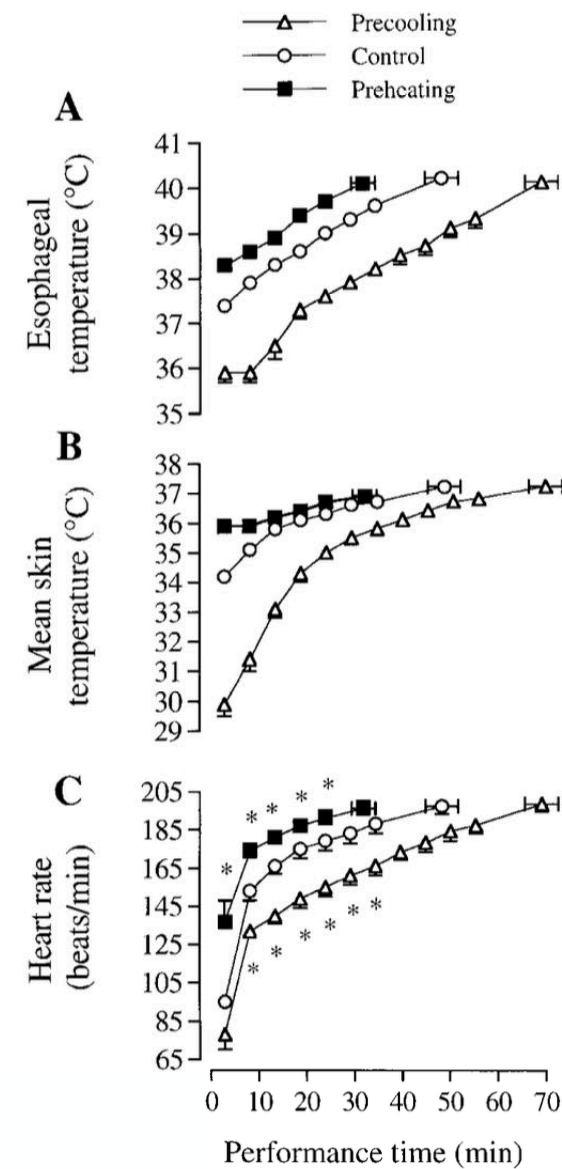
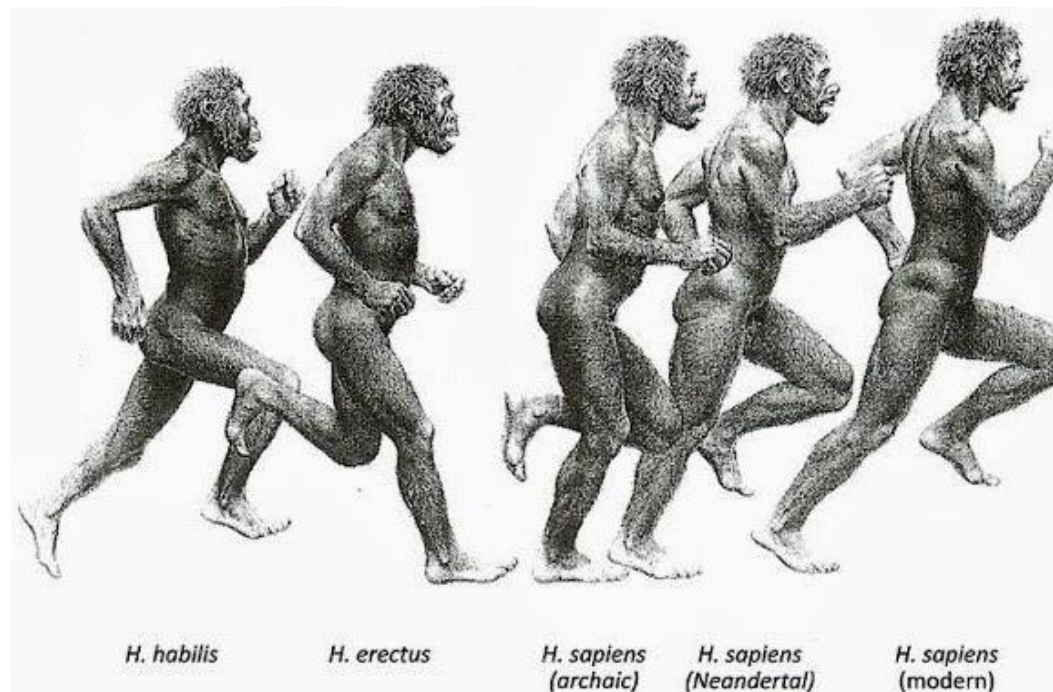


Chimpanzee genome project: **human skin is the organ that has undergone some of the most rapid evolutionary change since the split with chimpanzees**

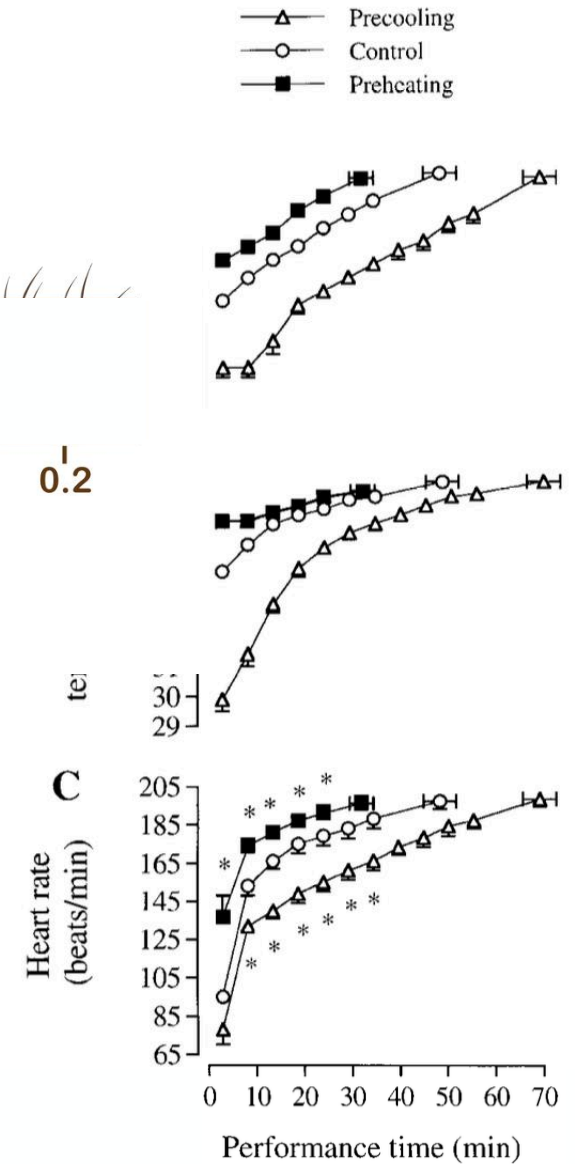
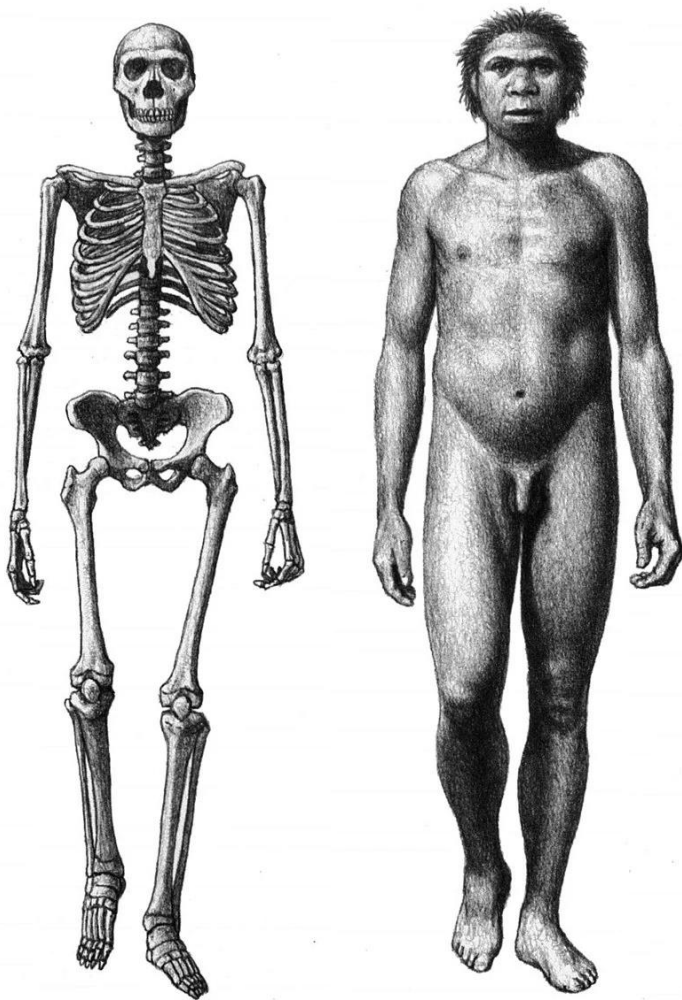
1. Why did humans *lose (most of)* their fur?



Turkana Boy (erectus)
Full modern human
proportion along distal femur to nailing



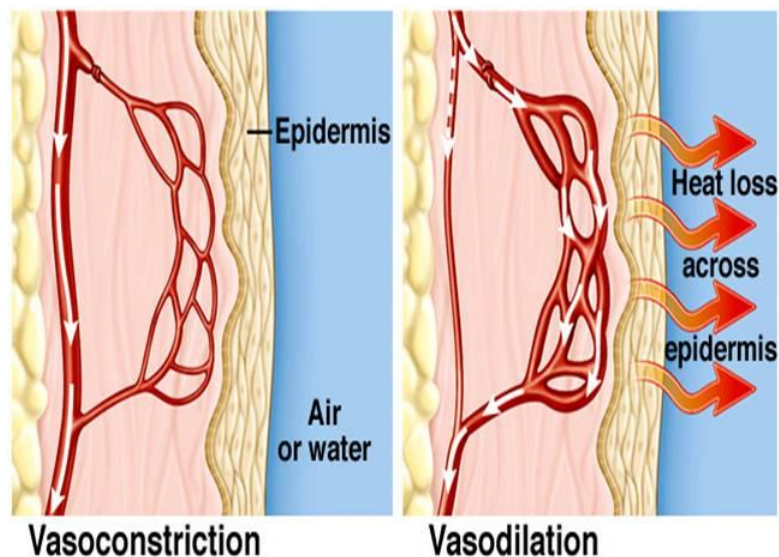
1. Why did humans *lose (most of)* their fur?



1. Why did humans *lose (most of)* their fur?

↳ Hominins

Human functional hairlessness is usually considered an adaptation for heat loss during extended bouts of locomotion



Surface area of skin directly exposed to air



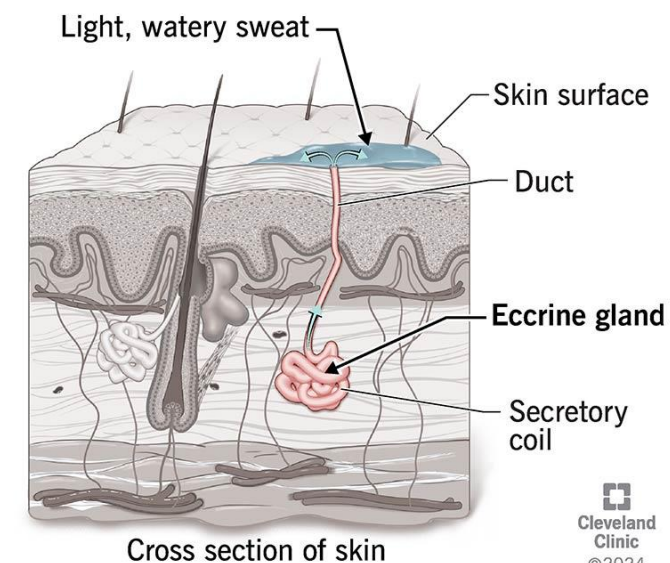
Lose insulation



Convective cooling

↳ Effective without fur but not w/ fur

Eccrine glands



Cleveland Clinic
©2024

Sweating: evaporative cooling

2. How is skin colour distributed across the world today? Why?

2. How is skin colour distributed across the world today? Why?

Melanin comes in two forms:



A dark pigment, usually brown or black in color

Focus of this lecture

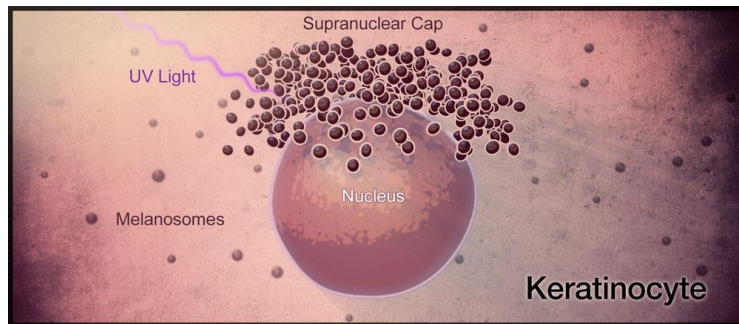


A lighter pigment, usually red or yellow in color

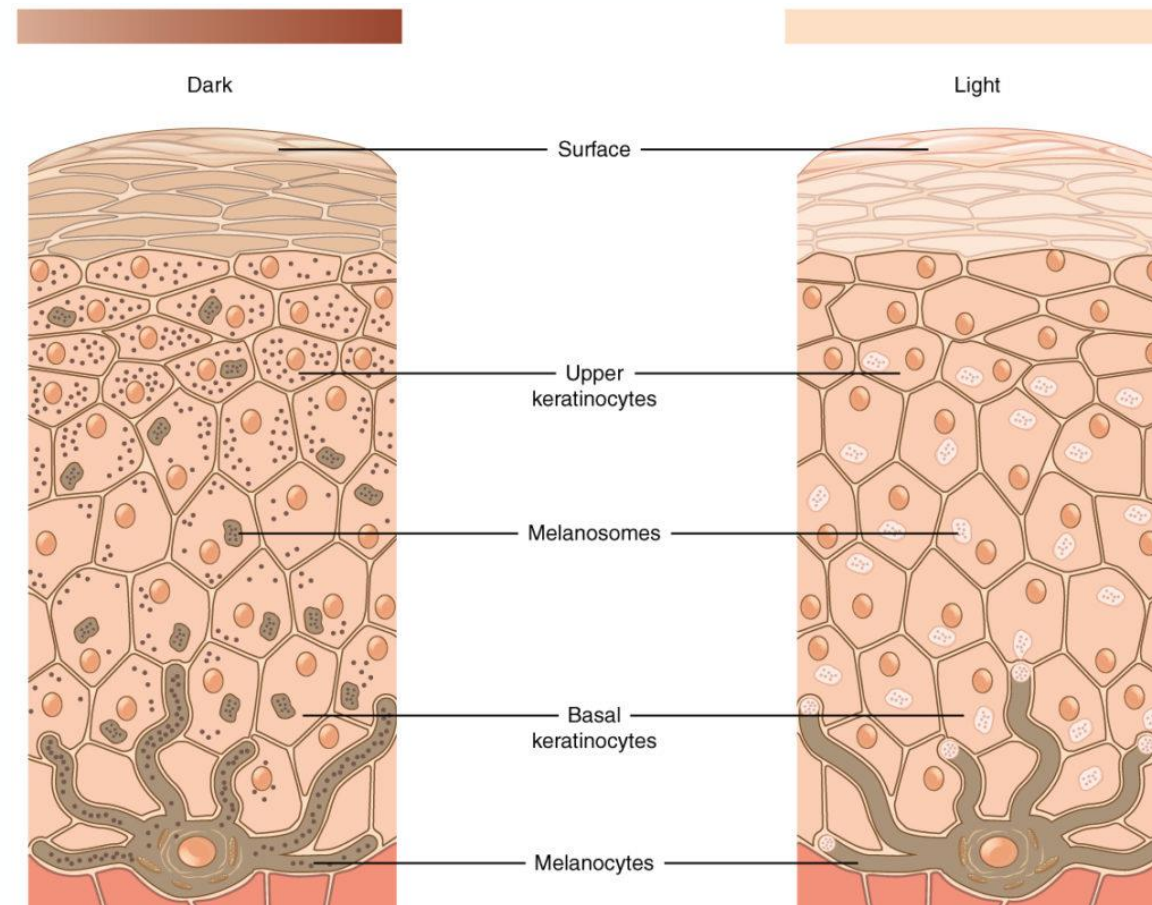
And is stored in membrane-bound sacs called melanosomes.

Melanin is stored

Melanosomes form a cap over the nucleus of skin cells.



Not randomly distributed in skin cells, form cap

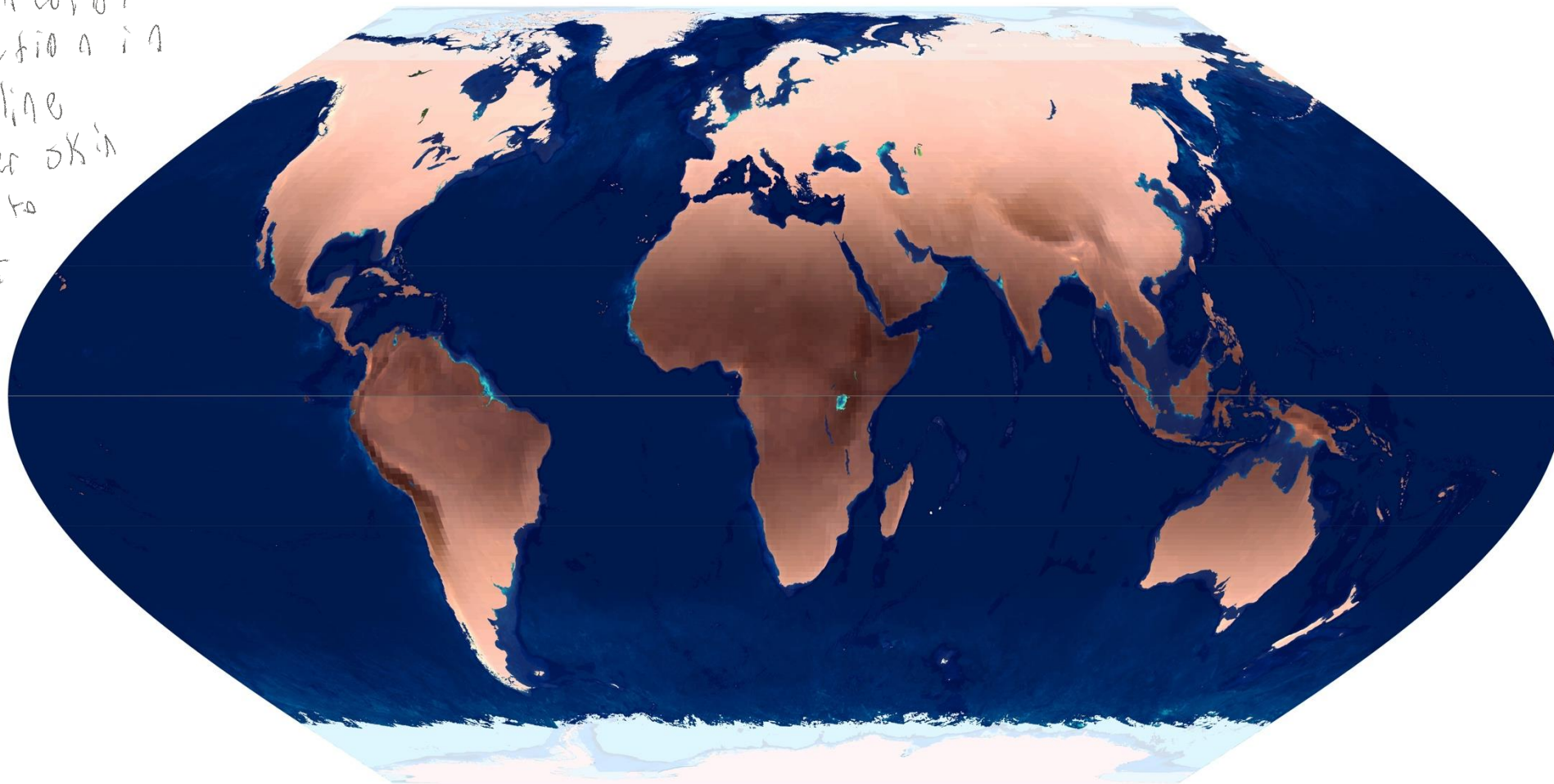


Melanosomes:

- Larger
- More numerous
- Distributed evenly

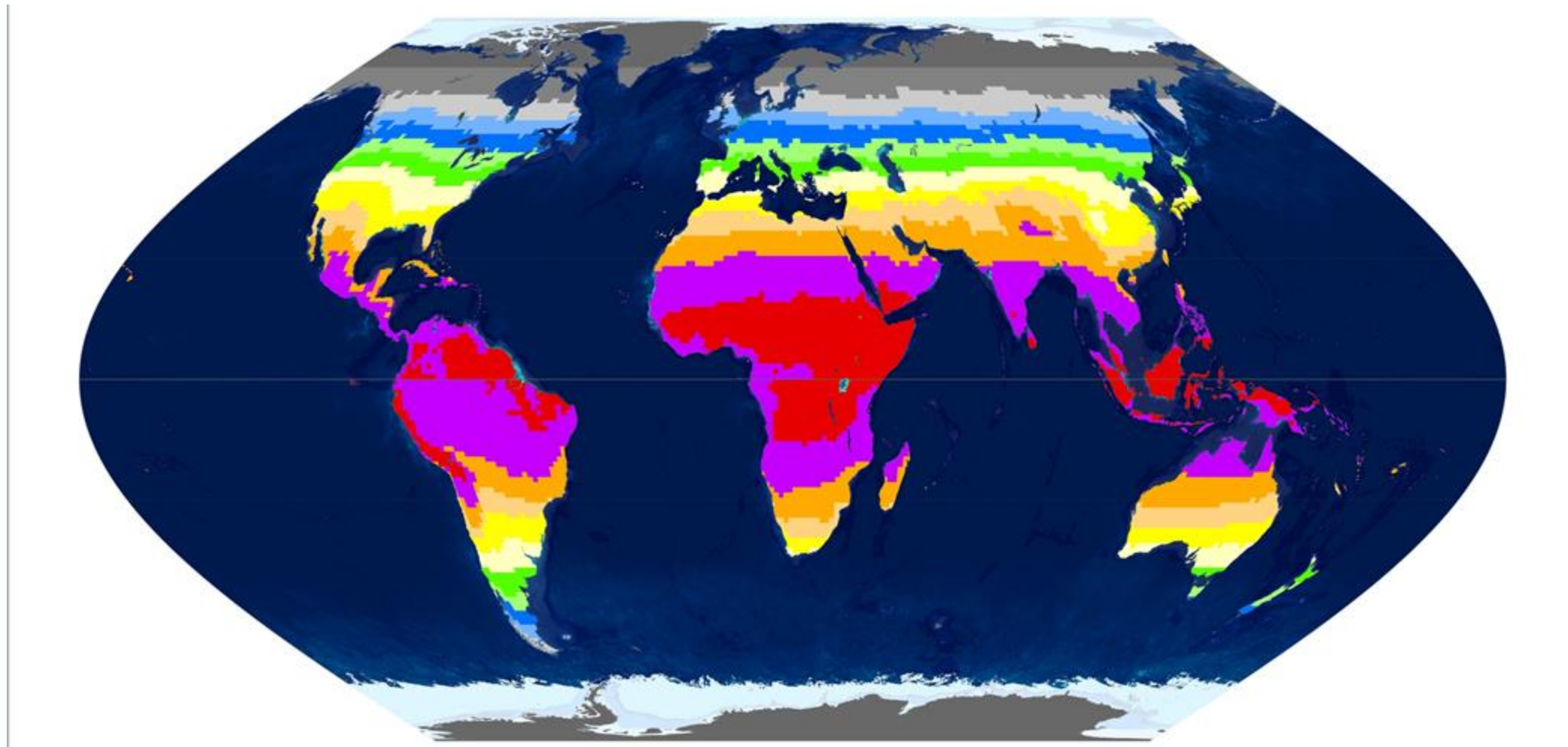
2. How is skin colour distributed across the world today? Why?

Clinical distribution
of skin color
+ direction in
this cline
↳ Darker skin
closer to
equator

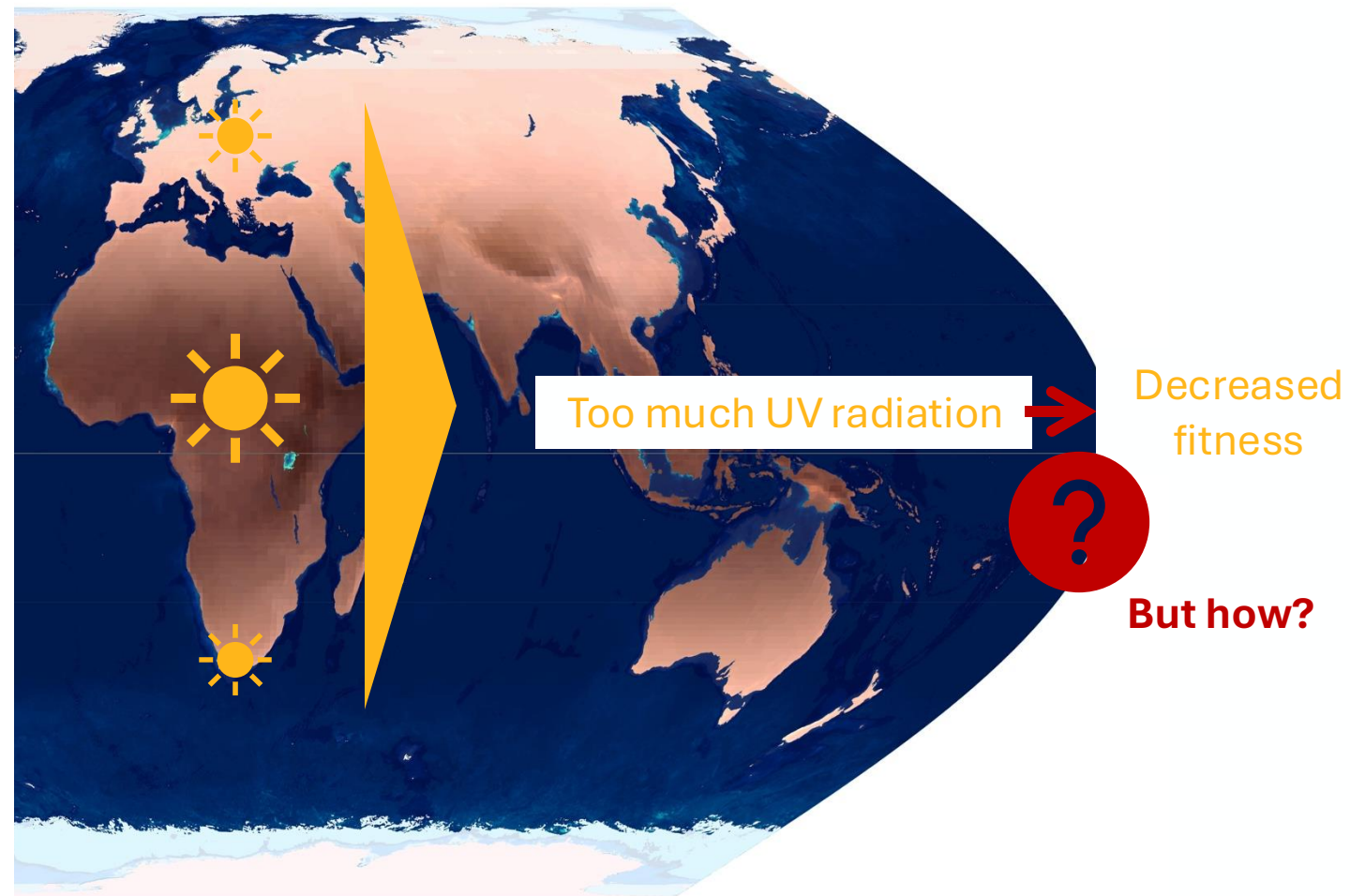
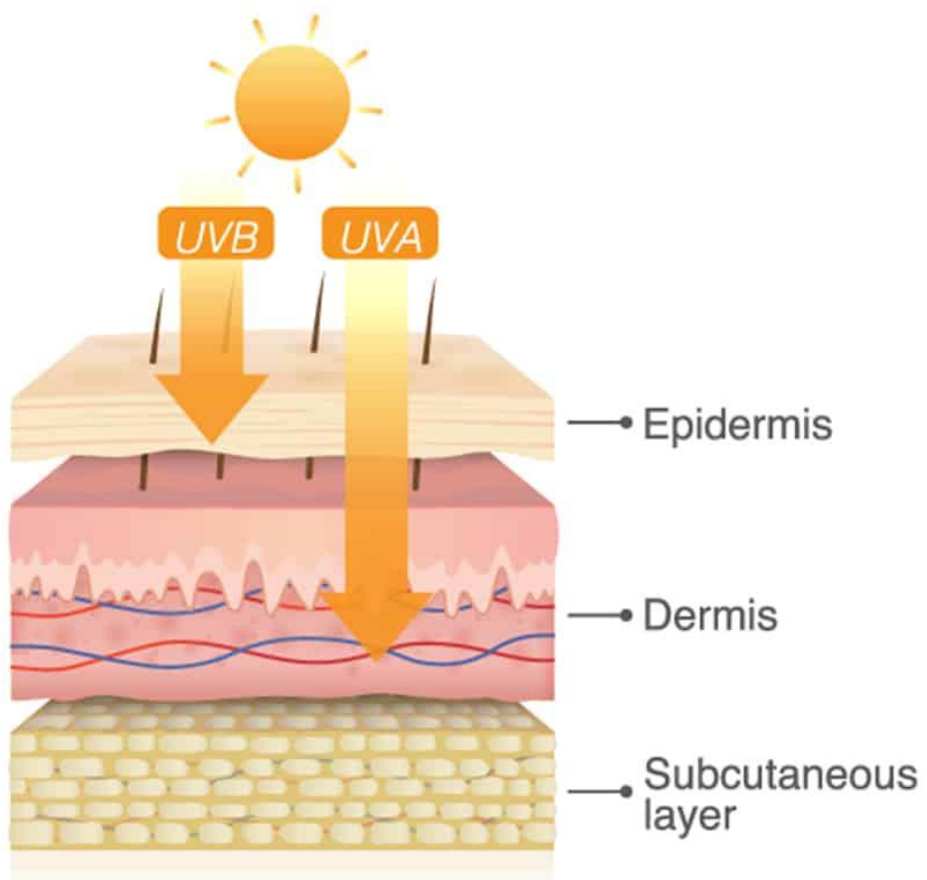


2. How is skin colour distributed across the world today? Why?

UV radiation intensity:

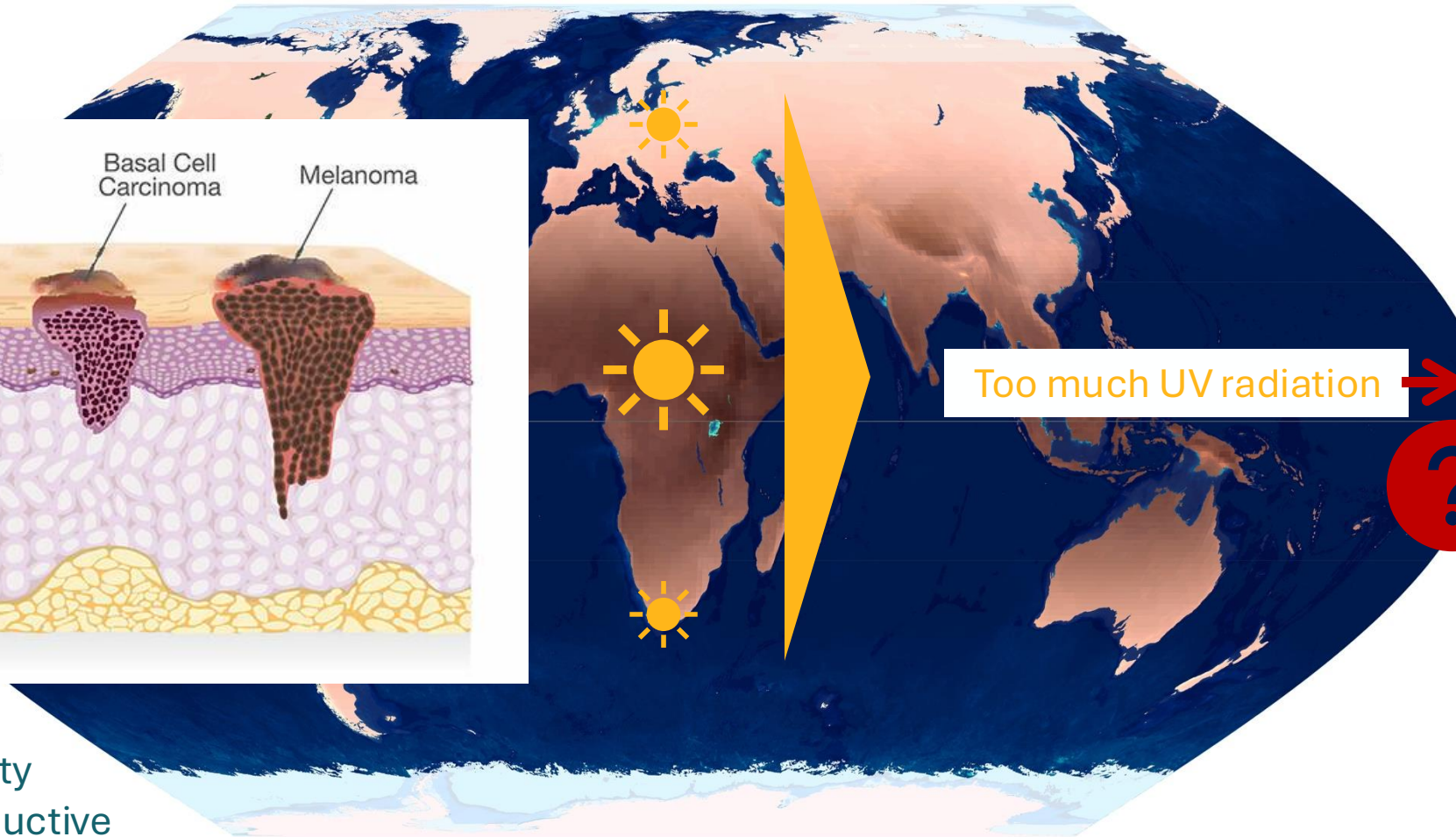
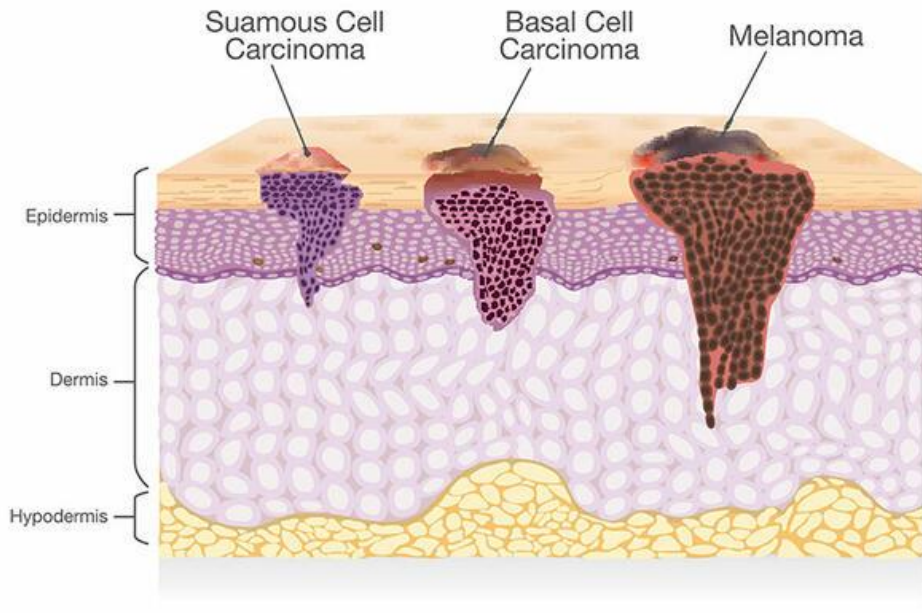


2. How is skin colour distributed across the world today? Why?



2. How is skin colour distributed across the world today? Why?

Hypothesis:



Decreased
fitness

But how?

But...

- Low mortality
- After reproductive career

tends to occur later in life

2. How is skin colour distributed across the world today? Why?

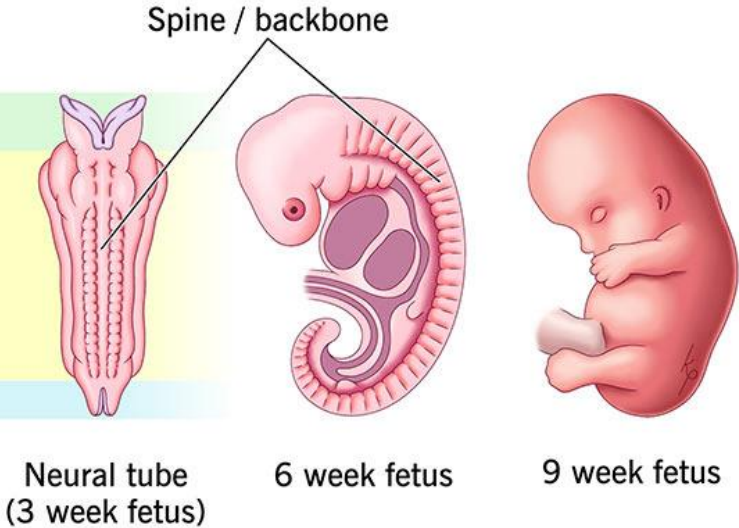
Neural Tube Defects (NTDs)

NTDs caused by location

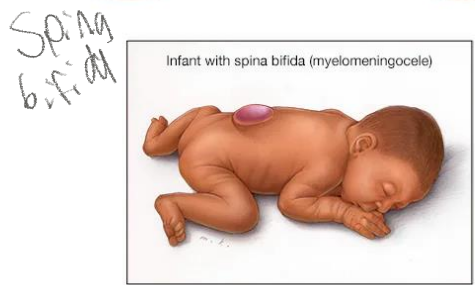
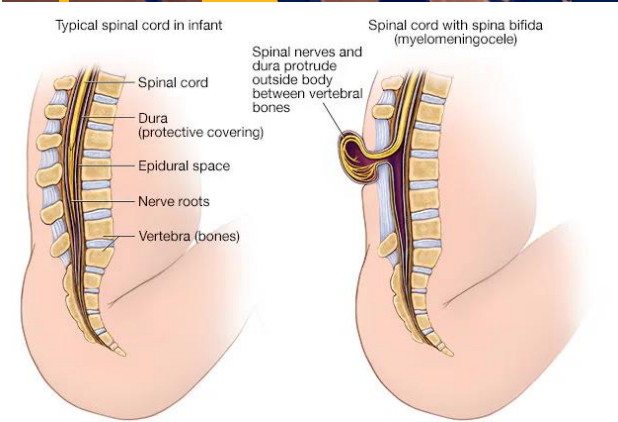
Anencephaly
Iniencephaly

Craniorachischisis
Encephalocele

Spina bifida



Cleveland Clinic ©2022



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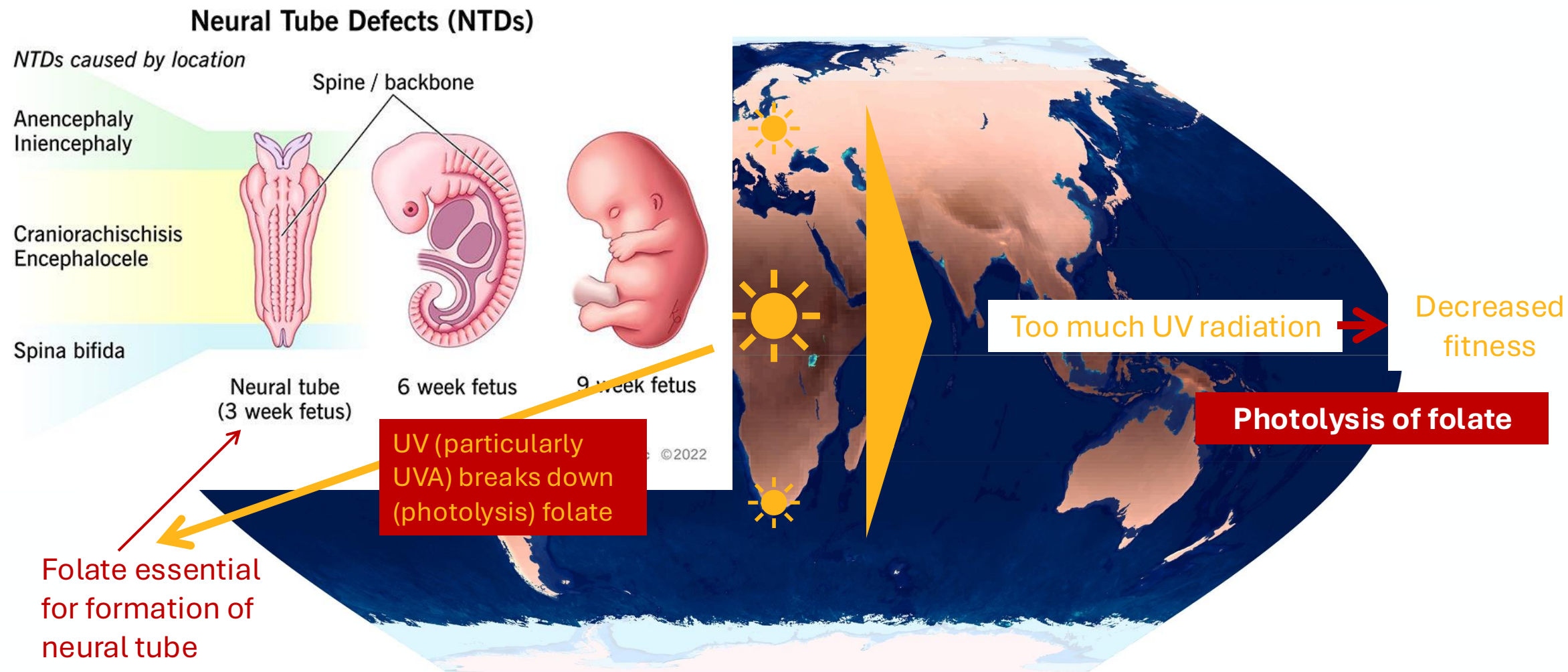
UV radiation

Decreased fitness

Photolysis of folate

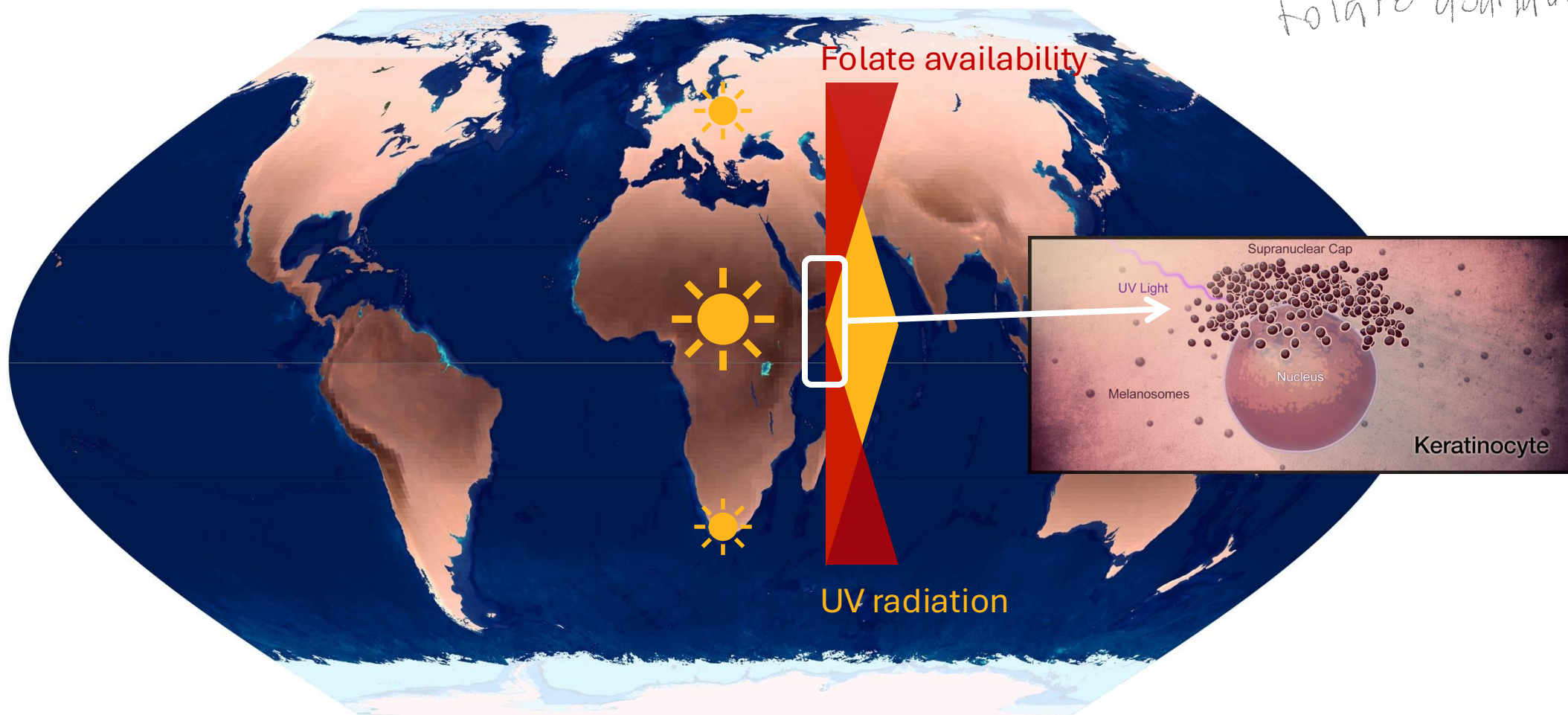
Folate essential for formation of neural tube

2. How is skin colour distributed across the world today? Why?

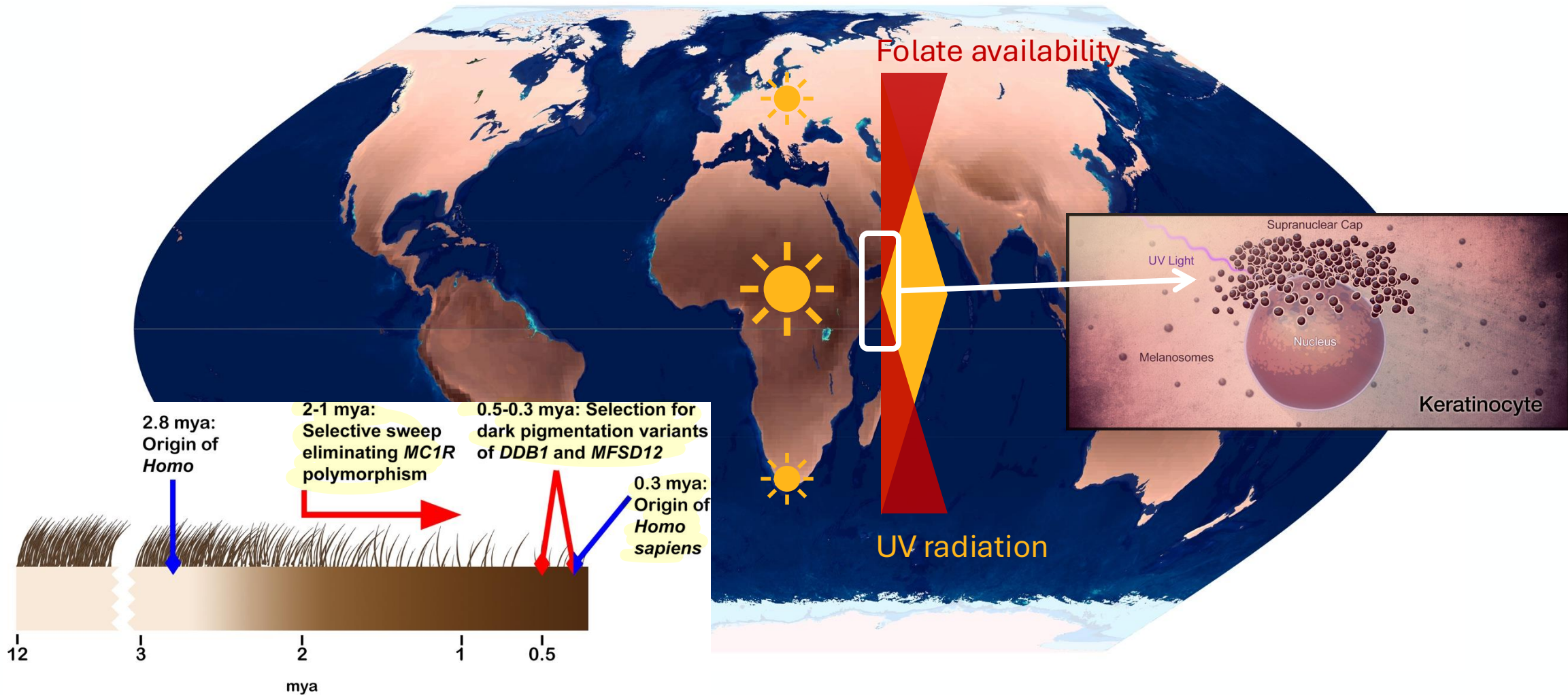


2. How is skin colour distributed across the world today? Why?

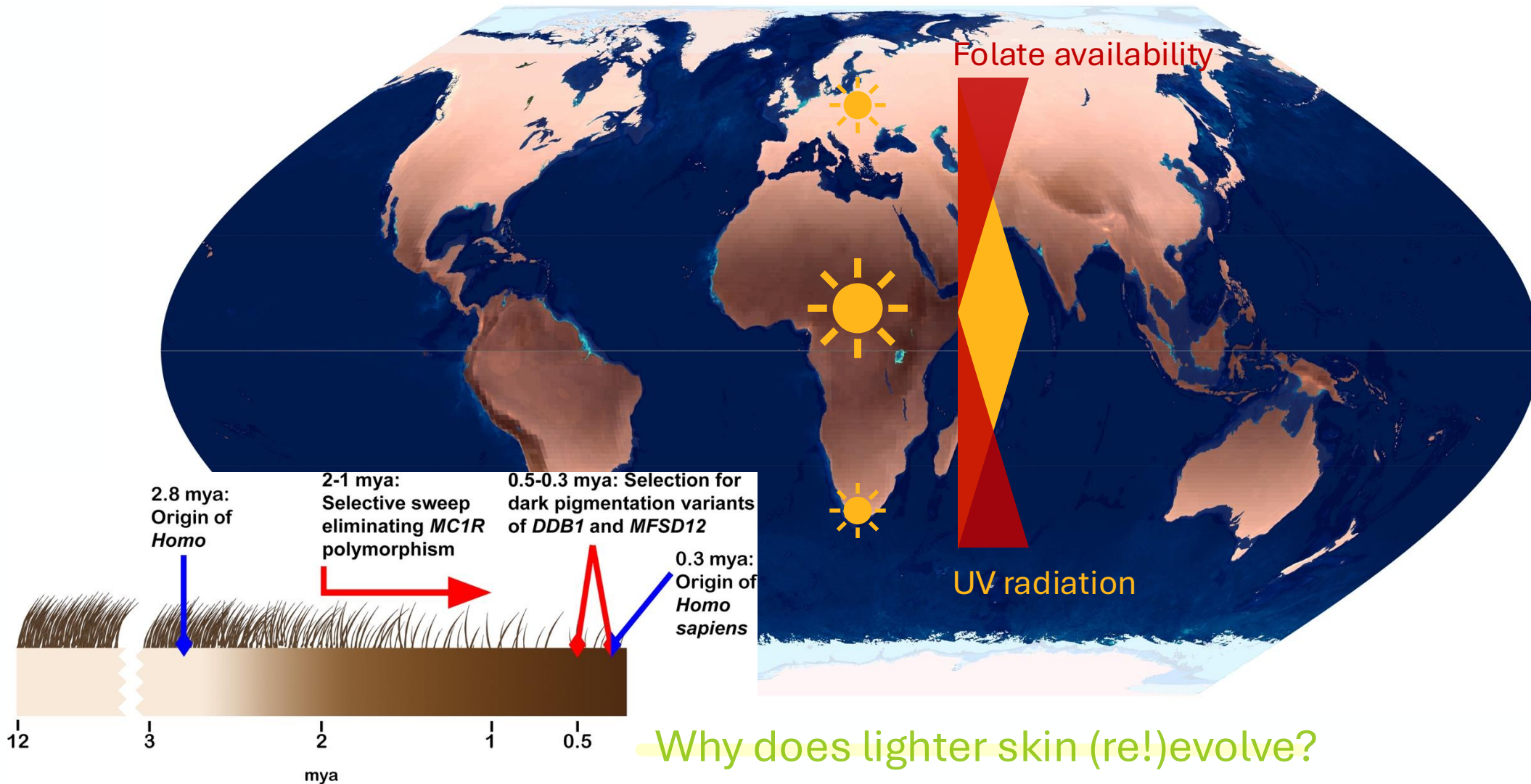
low UV radiation $\rightarrow$ high
folate availability



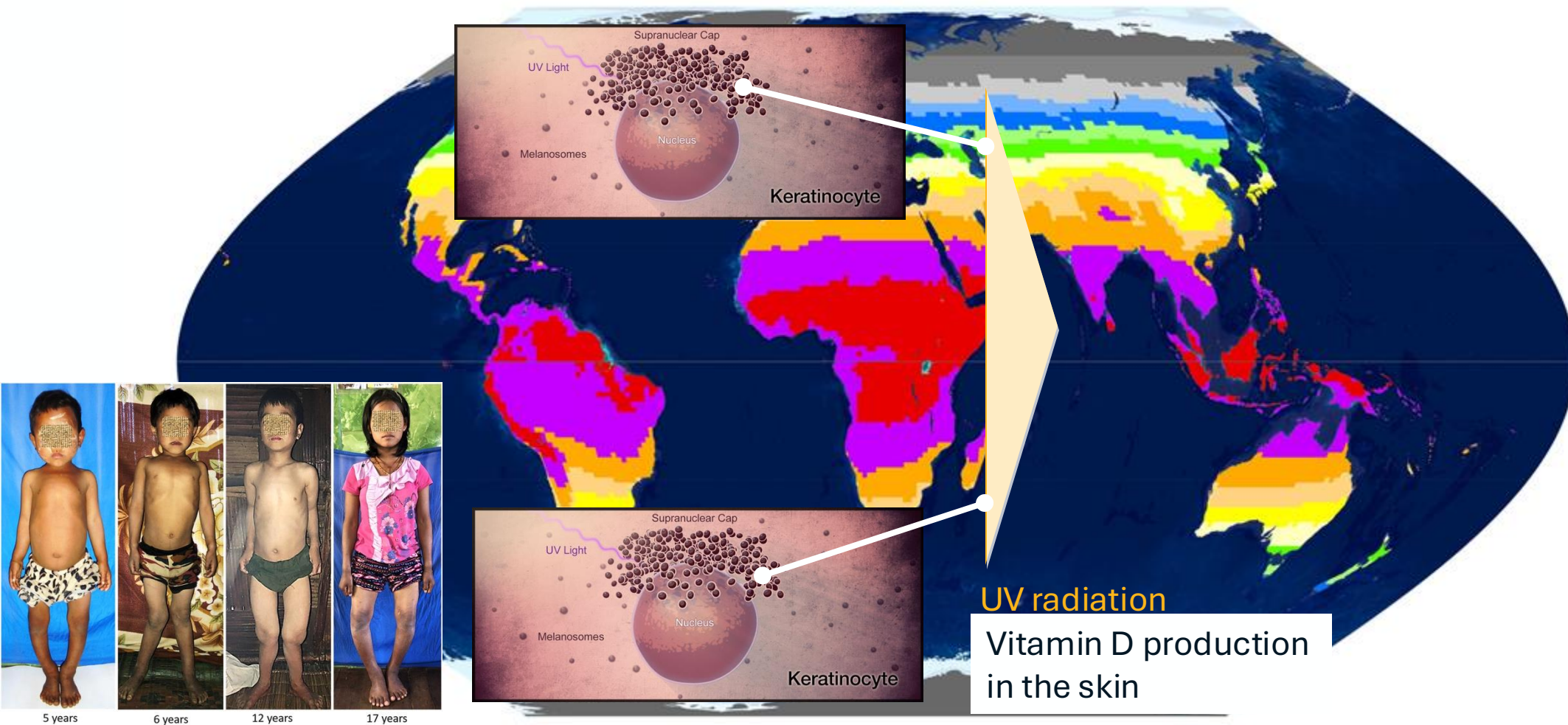
2. How is skin colour distributed across the world today? Why?



2. How is skin colour distributed across the world today? Why?



2. How is skin colour distributed across the world today? Why?



Vitamin D deficiency: rickets, but also pelvic shape abnormalities

2. How is skin colour distributed across the world today? Why?

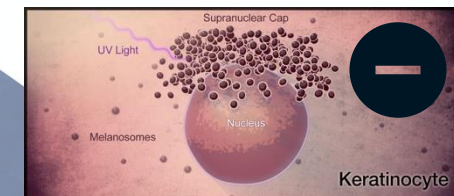
The clinal pattern we see w/ skin color reflects tradeoffs in UV penetration

Latitudinal gradient reflects **tradeoffs** in how much UV radiation penetrates into the skin

Low UV radiation

- Low rate of folate photolysis
- High chance of vitamin D deficiency

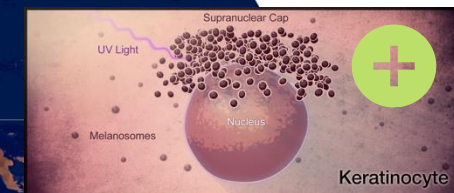
Selection for decreased anti-UV protection



High UV radiation

- High rate of folate photolysis
- Low chance of vitamin D deficiency

Selection for increased anti-UV protection



Measuring requires description... but how good are our descriptions?

Fitzpatrick Scale ^{typical skin color scale}

					
Type I	Type II	Type III	Type IV	Type V	Type VI
Light, Pale White	White, Fair	Medium White to Olive	Olive Tone	Light Brown	Dark Brown
Always burns, never tans	Usually burns, tans with difficulty	Sometimes mild burn, gradually tans to olive	Rarely burns, tans with ease to moderate brown	Very rarely burns, tans very easily	Never burns, tans very easily, deeply pigmented

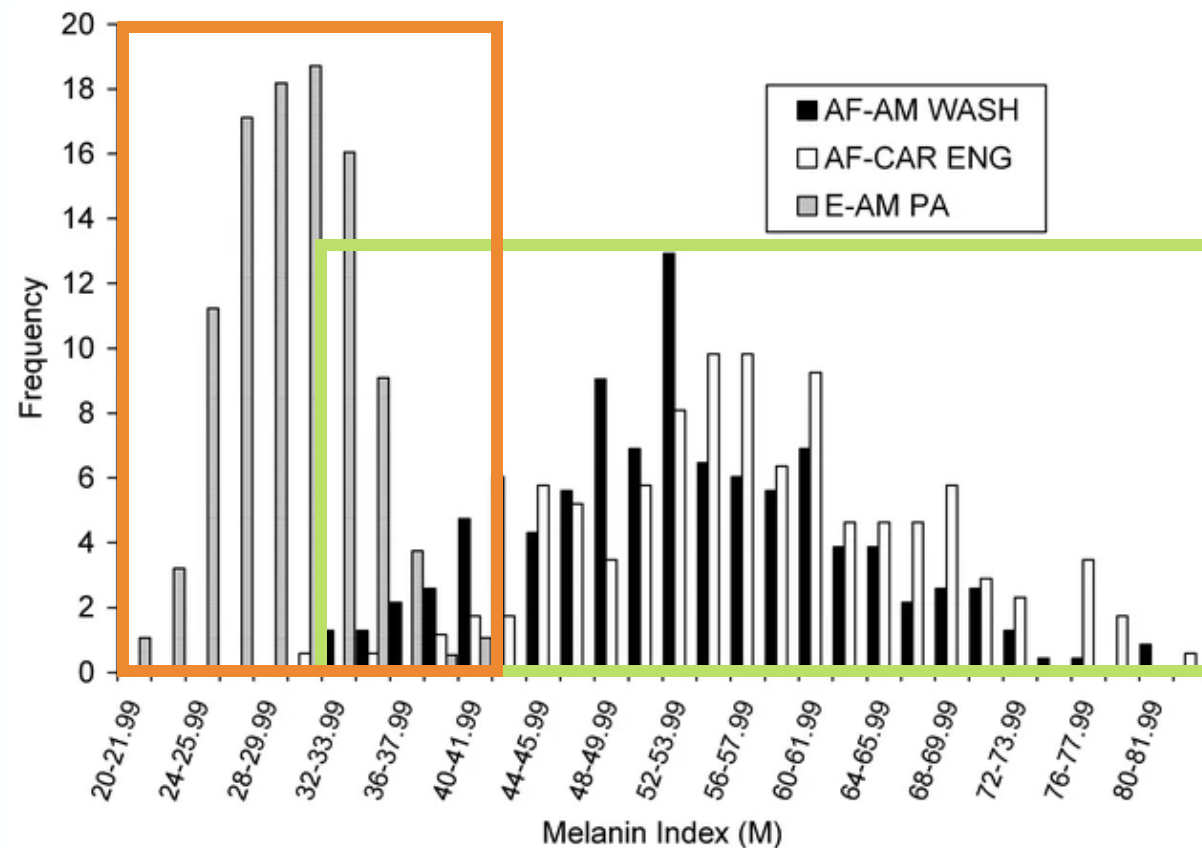
European descent

African descent

We want a good category to capture the same amount of variation in each type. Should capture an even % of variation.

Is the amount of variation seen in people of European descent objectively 2x bigger than that seen in people of African descent?

Measuring requires description... but how good are our descriptions?



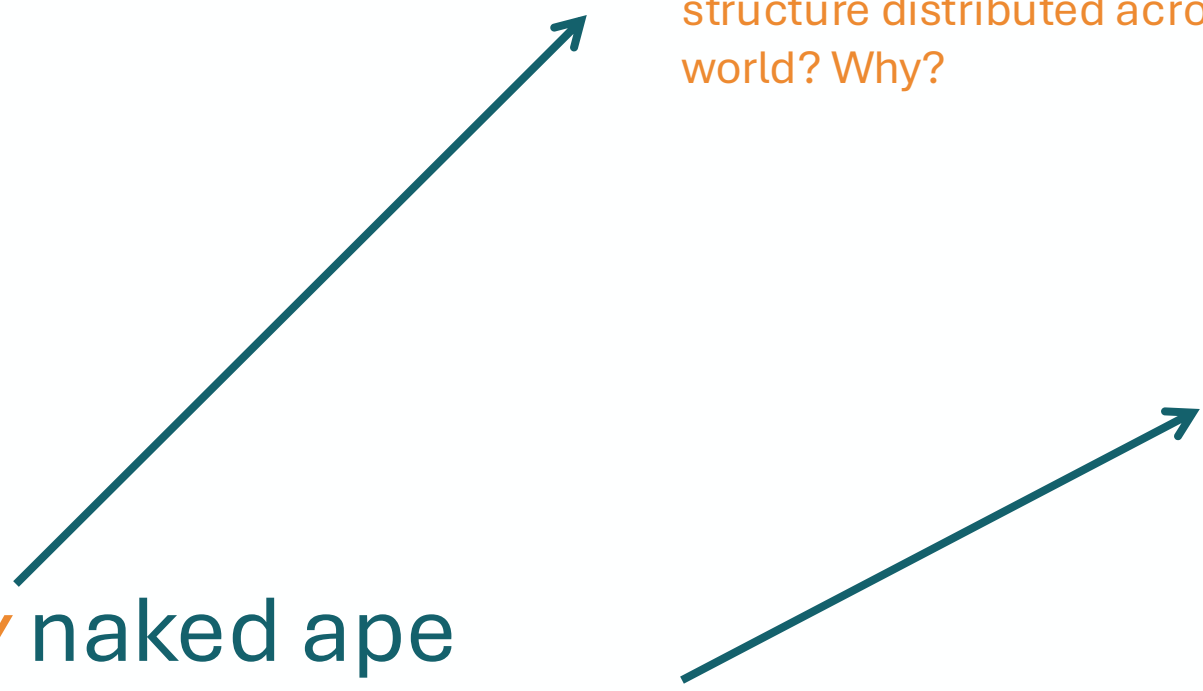
Is the amount of variation seen in people of European descent objectively 2x bigger than that seen in people of African descent?

NO

- Why did humans lose their fur?
- Why do humans have scalp hair that is so exceptionally diverse in colour *and* structure?
- How is scalp hair colour and structure distributed across the world? Why?

- Why did humans lose their fur?
- Why do humans have skin that is so exceptionally diverse in colour?
- How is skin colour distributed across the world? Why?

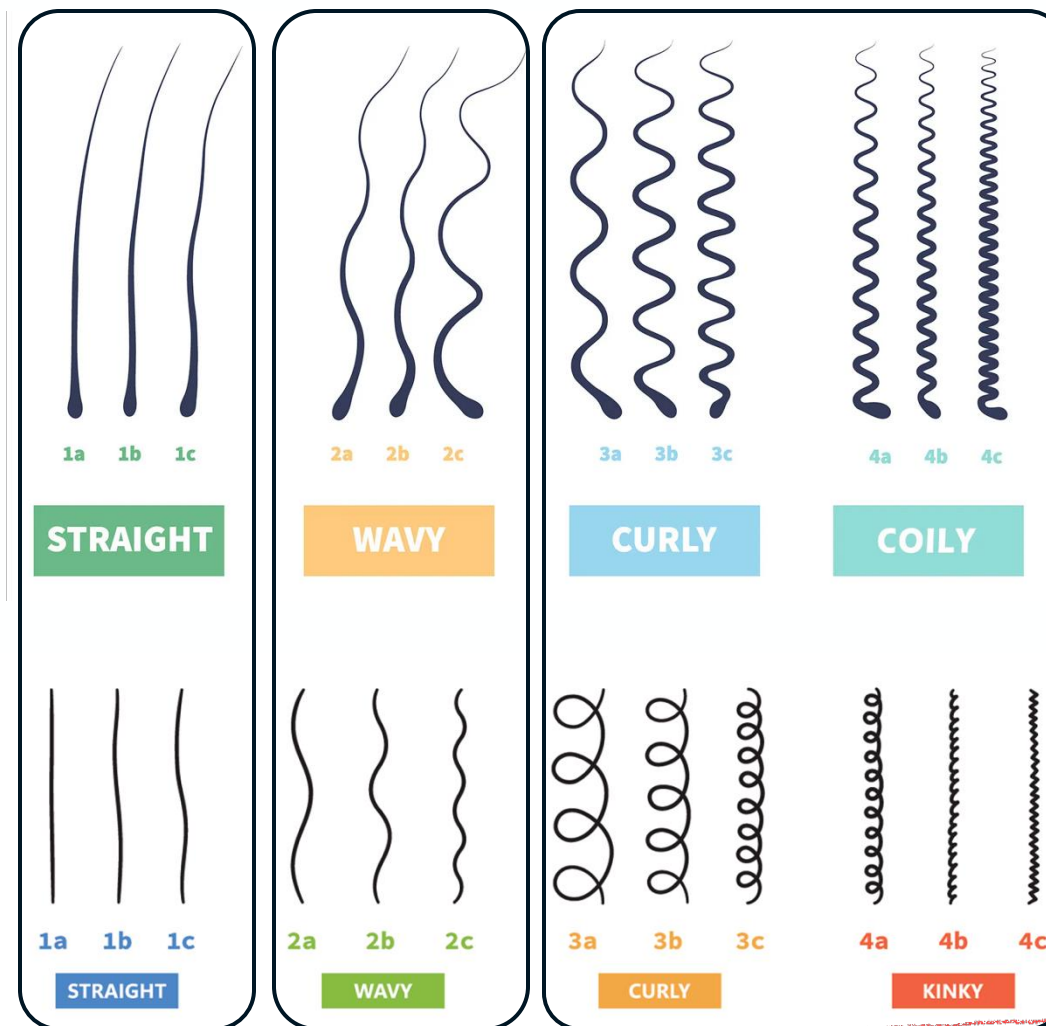
The *nearly* naked ape
with exceptional skin colour
diversity



1. How is hair form distributed across the world today? Why?

How are we describing variation?

Are our categories capturing equivalent amounts of variation?



5 out of 8 studies

↳ 3 point to system
btwn straight - wavy - curly

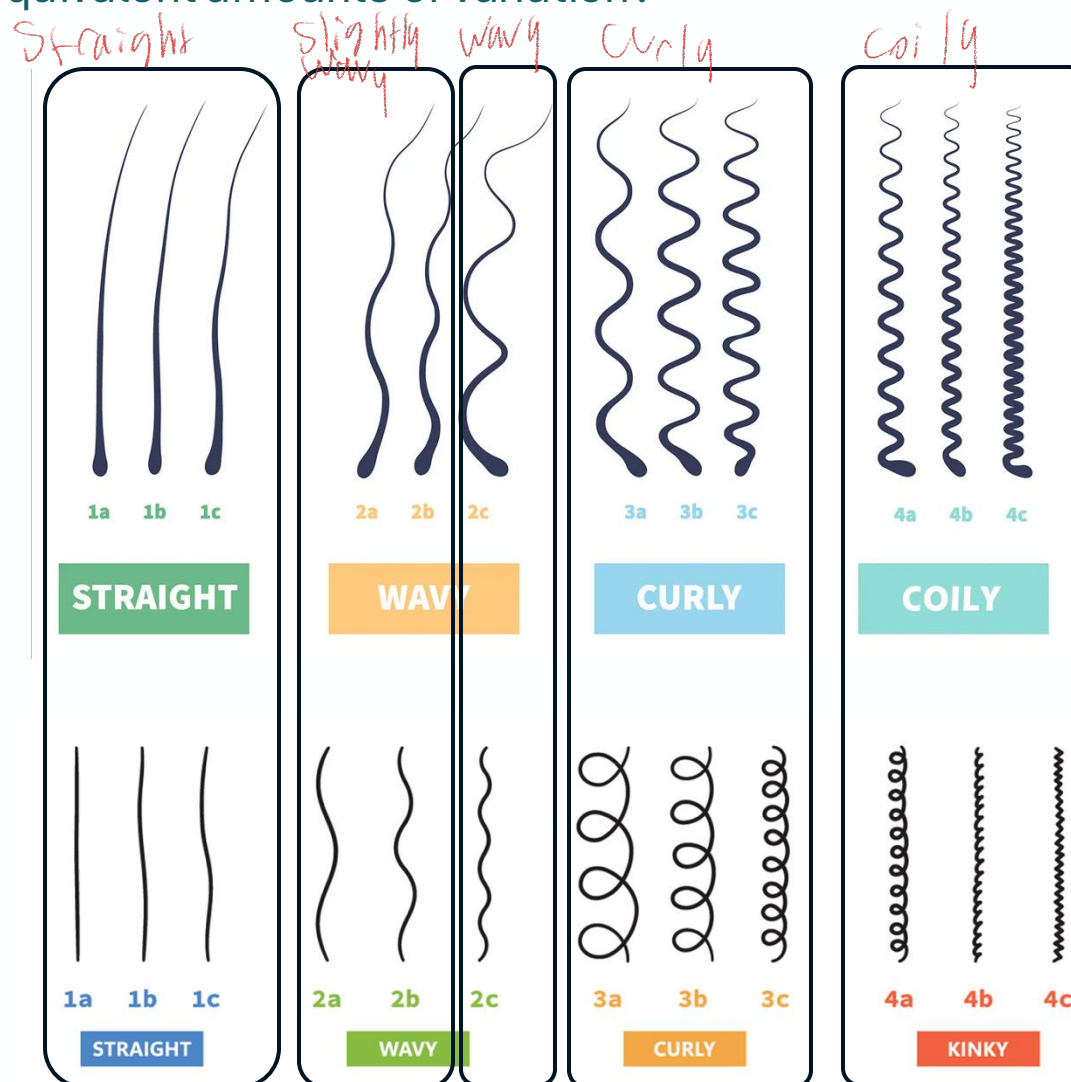
are differences
btwn 1a and 1b equivalent
to 2a and 2b?

everything is
curlier
than wavy
hair

1. How is hair form distributed across the world today? Why?

How are we describing variation?

Are our categories capturing equivalent amounts of variation?

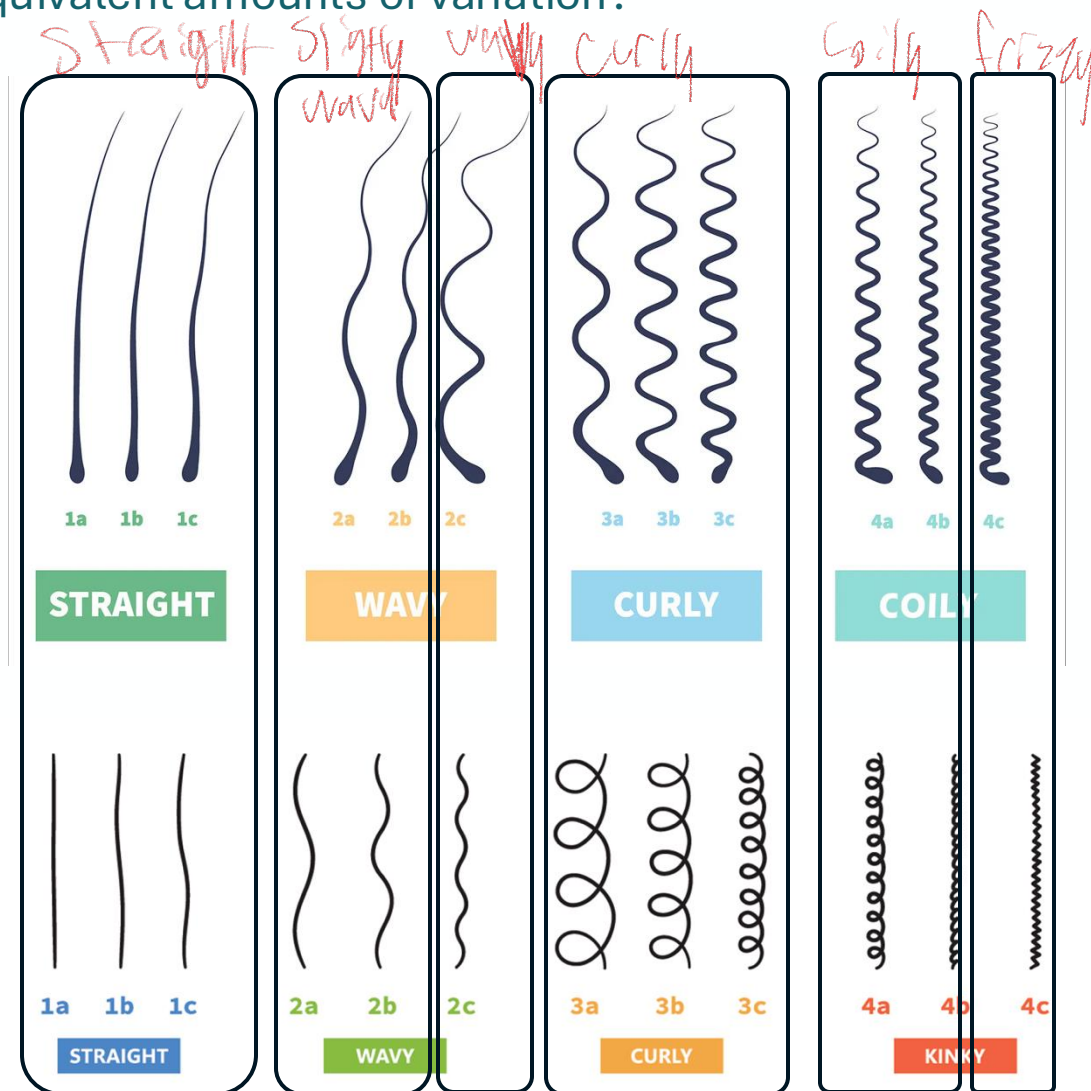


1 out of 8 studies

1. How is hair form distributed across the world today? Why?

How are we describing variation?

Are our categories capturing equivalent amounts of variation?

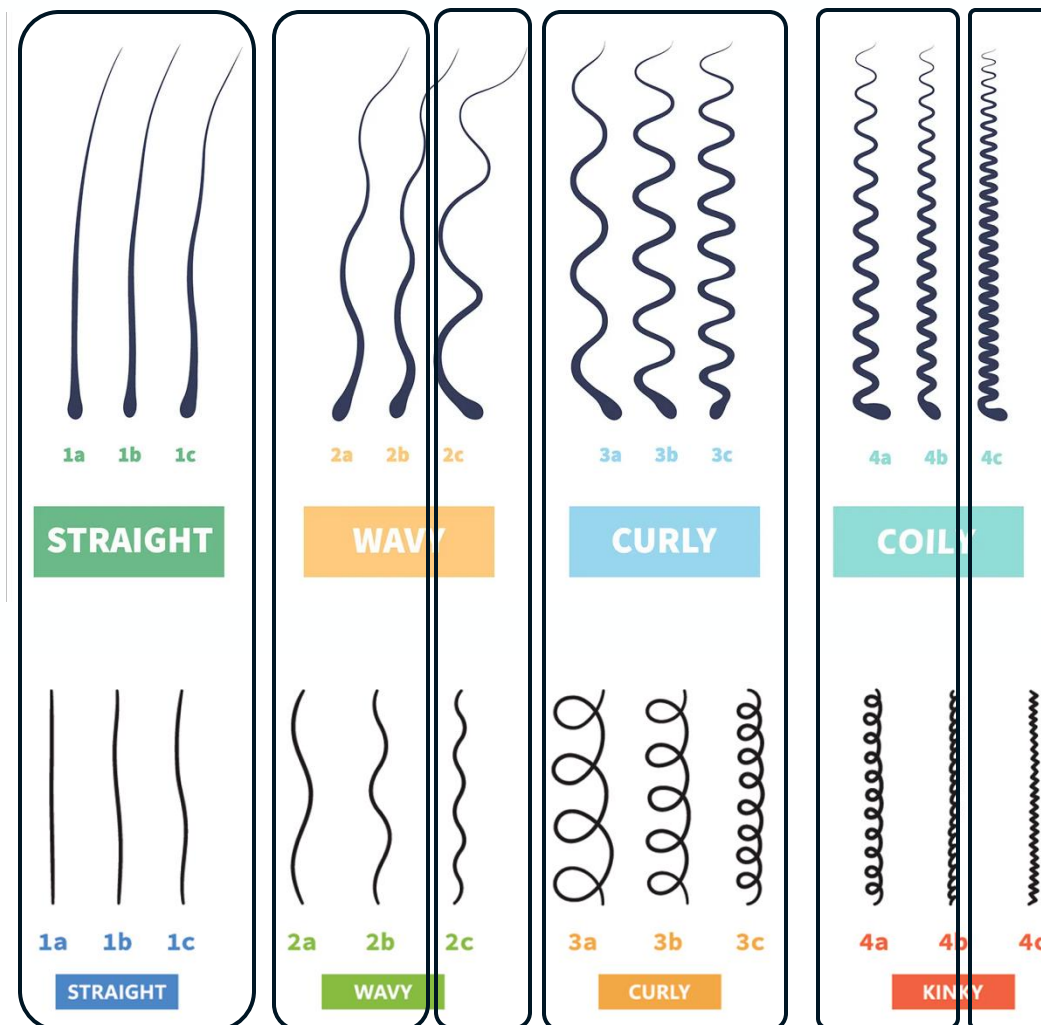


2 out of 8 studies

1. How is hair form distributed across the world today? Why?

How are we describing variation?

Are our categories capturing equivalent amounts of variation?



2 out of 8 studies

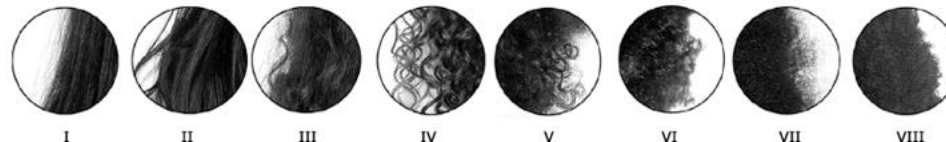
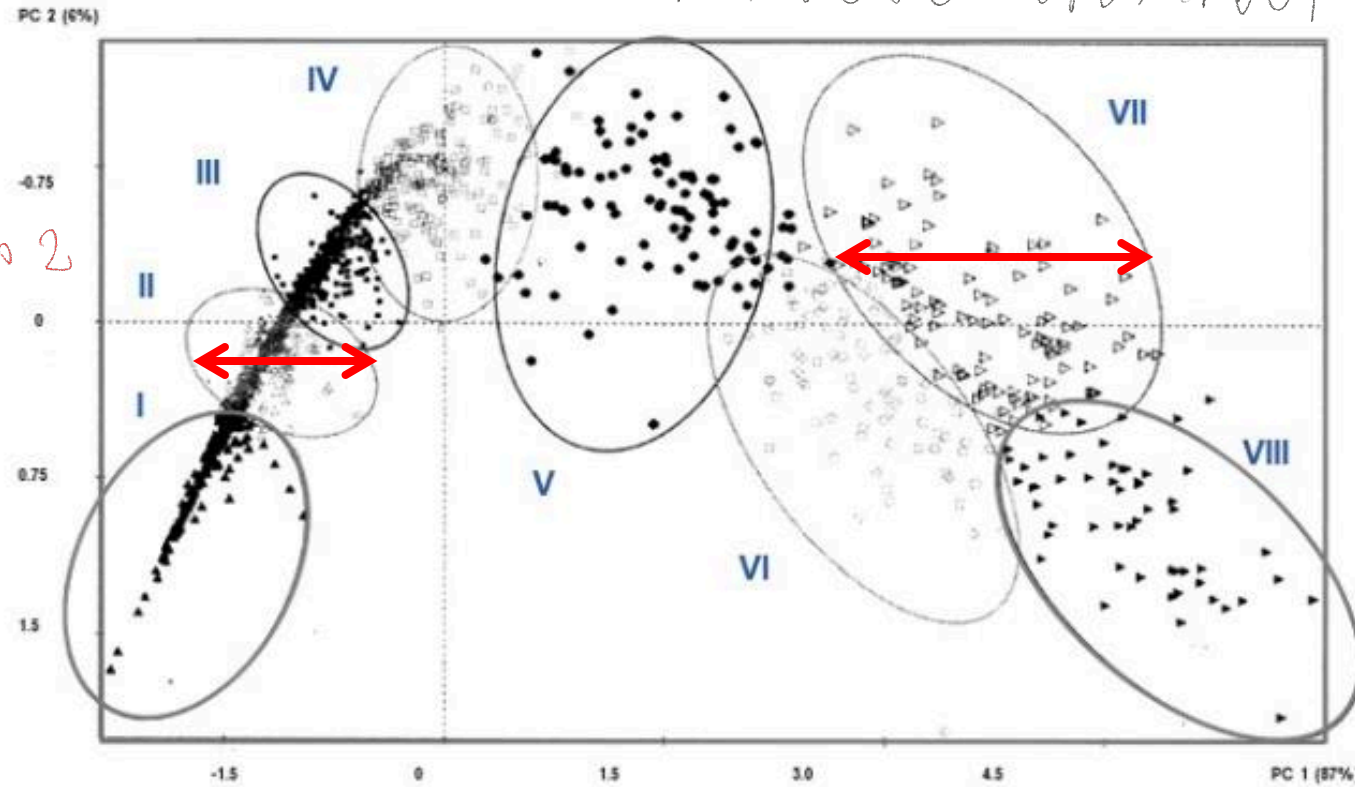
+ what about cross-section (e.g., thickness)?

1. How is hair form distributed across the world today? Why?

continuous distribution -

but these categories are not distributed well

PCA -
all of these
variables,
reducing down to 2
PCs describe
most diversity
PC 2 describes
second most
diversity



1. How is hair form distributed across the world today? Why?

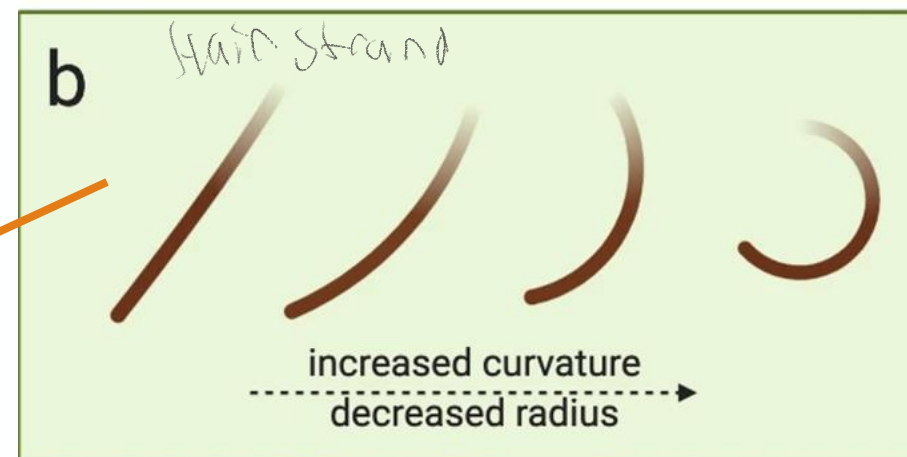
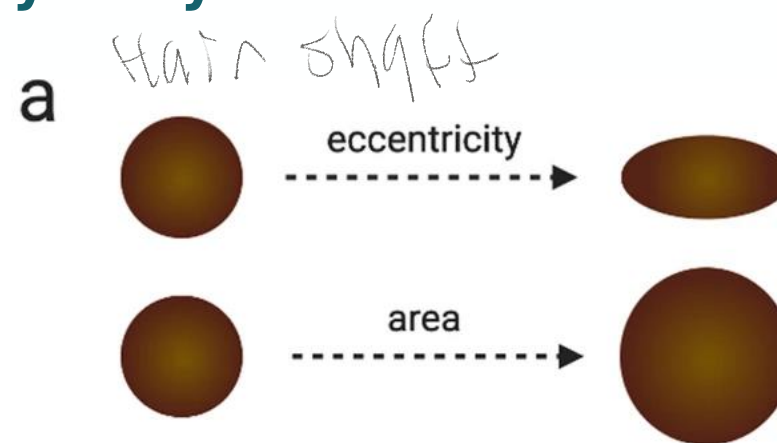
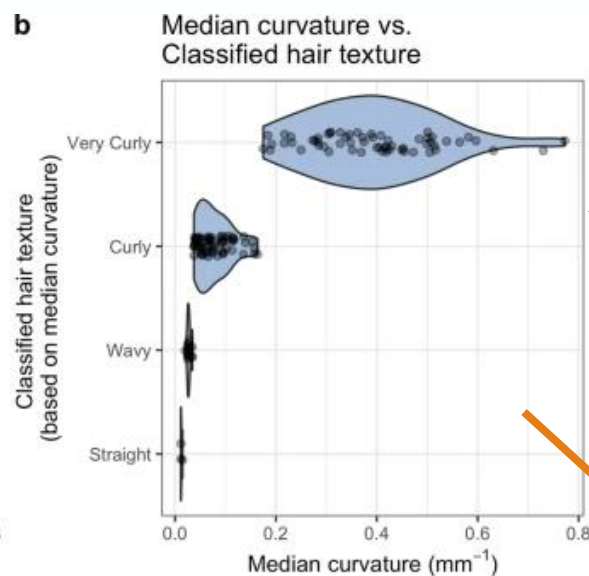
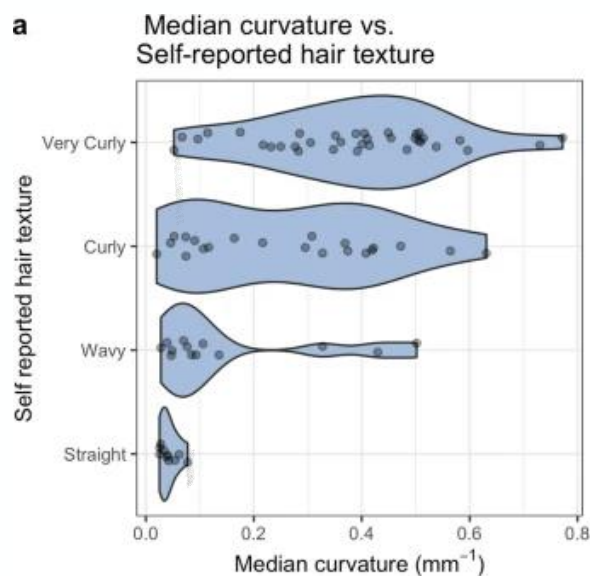
Article | [Open access](#) | Published: 01 June 2021

High-throughput phenotyping methods for quantifying hair fiber morphology

[Tina Lasisi](#) , [Arslan A. Zaidi](#), [Timothy H. Webster](#), [Nicholas B. Stephens](#), [Kendall Routh](#), [Nina G. Jablonski](#) & [Mark D. Shriver](#)

[Scientific Reports](#) **11**, Article number: 11535 (2021) | [Cite this article](#)

7954 Accesses | 8 Citations | 81 Altmetric | [Metrics](#)



Self-reported hair texture and “objective” classification fail to capture quantitative variation in curvature

1. How is hair form distributed across the world today? Why?

We don't have
good measurements of
hair form →
No figure of
distribution

1. Ok... then at least tell me why we retained scalp hair, and why there is variation in its form.

Curls keep you cool!

Tightly curled hair offers greater protection from solar radiation heat

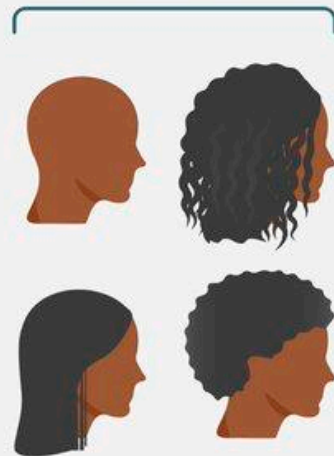
We came to this conclusion by running a series of experiments

We used lamps to simulate solar radiation



We used a thermal manikin to measure heat transfer

We put wigs with different textures on the manikin



We ran the experiments with different wind speeds and with dry and wet scalps (to simulate sweat)



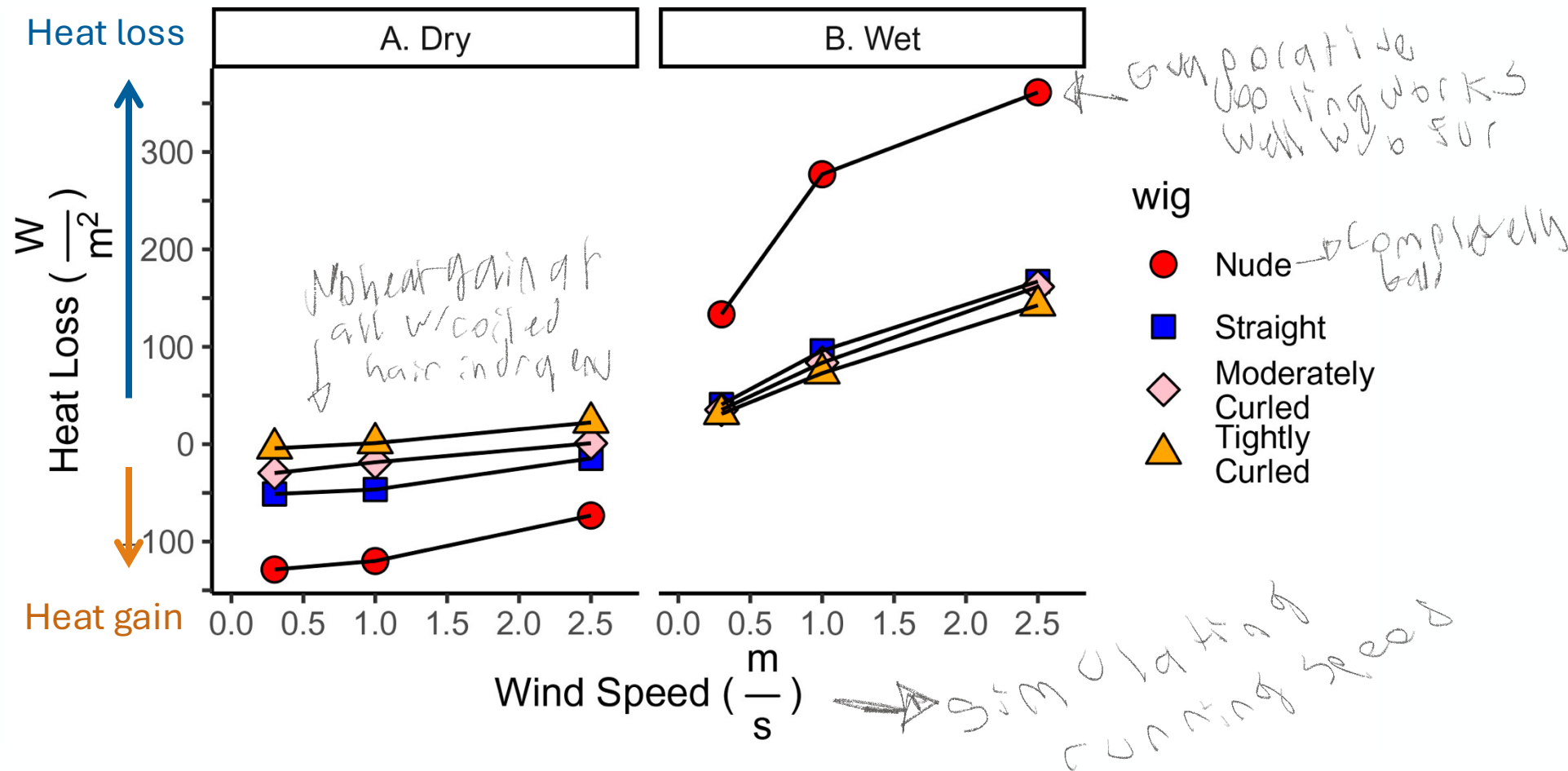
For each combination of factors we measured heat transfer from the scalp

Our results likely reflect the fact that tightly curled hair creates more distance between the scalp and the top of the hair.

1. Ok... then at least tell me why we retained scalp hair, and why there is variation in its form.



1. Ok... then at least tell me why we retained scalp hair, and why there is variation in its form.



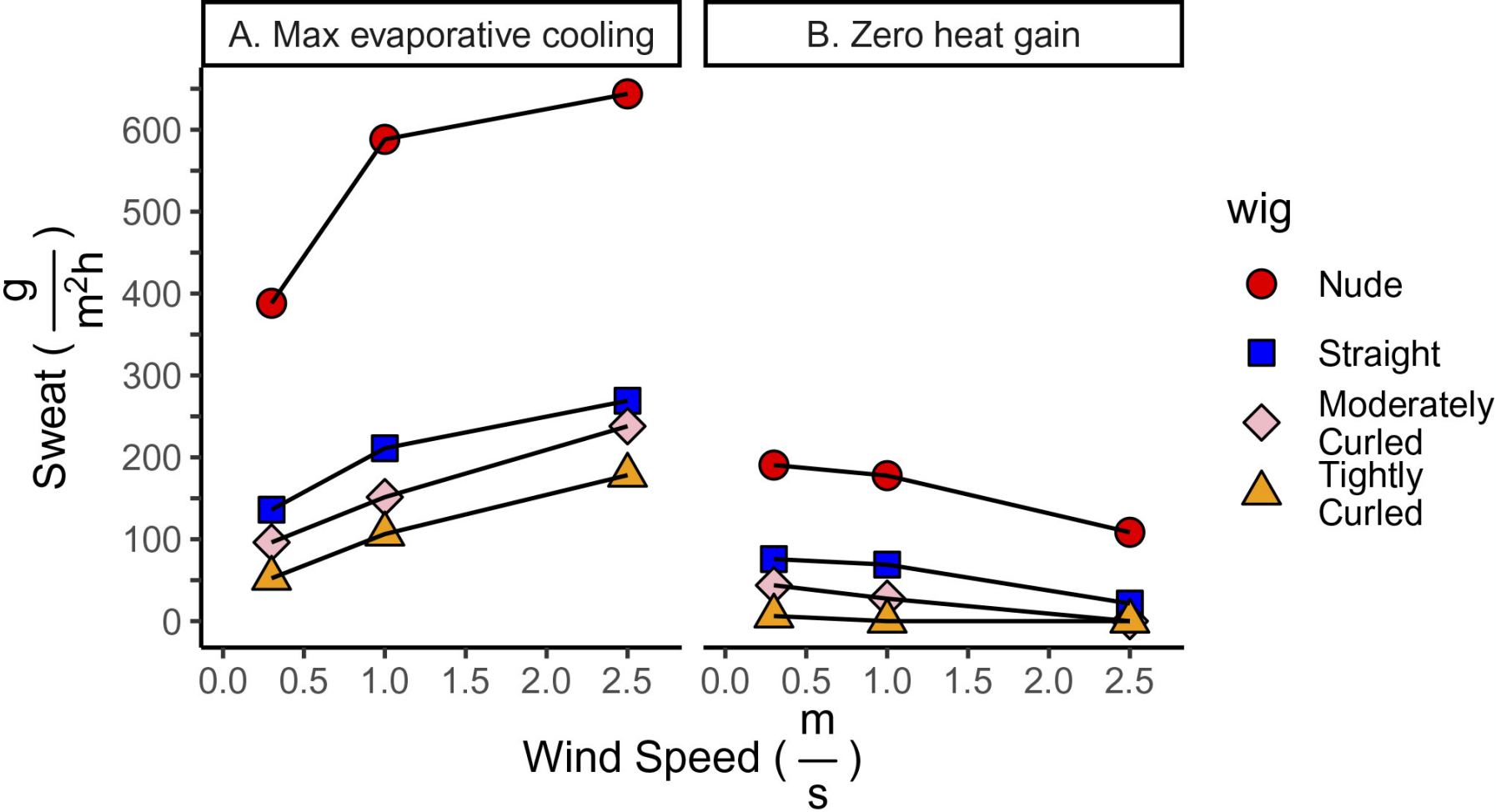
Curls reduce heat gain in dry context

Don't need to sweat onto @ of head w/ curls → less dehydration

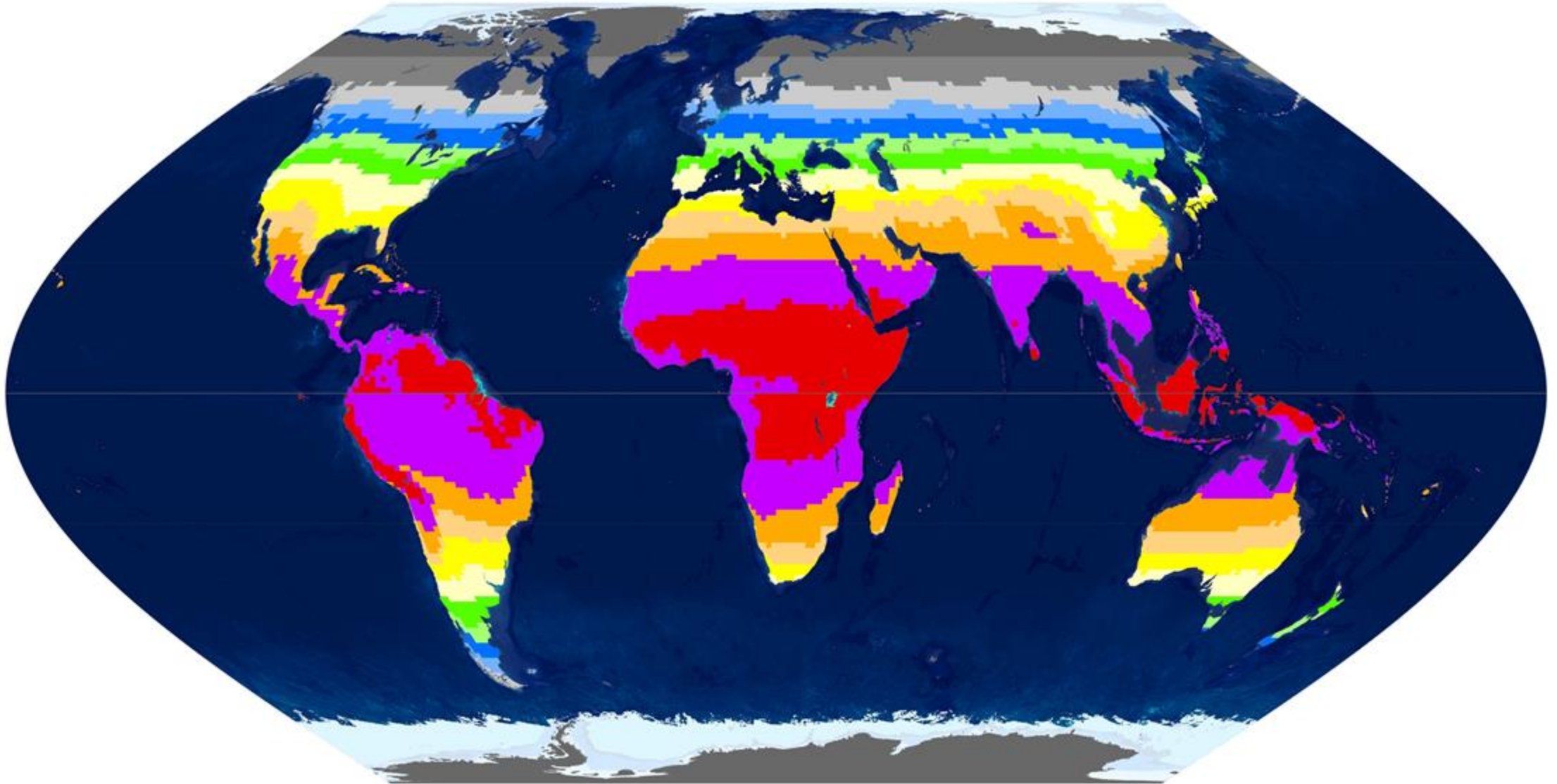
Tight curls require least sweat because they prevent heat gain

Key

Estimated required sweat rate to achieve



Making predictions...



- Why did humans lose their fur?
- Why do humans have scalp hair that is so exceptionally diverse in colour *and* structure?
- How is scalp hair colour and structure distributed across the world? Why?

Running / heat stress

- Why did humans lose their fur?
- Why do humans have skin that is so exceptionally diverse in colour?
- How is skin colour distributed across the world? Why?

PP can be by of 1 so heat stress

The *nearly* naked ape
with exceptional skin colour
diversity